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Part 1

1. (a) Complete the following table:

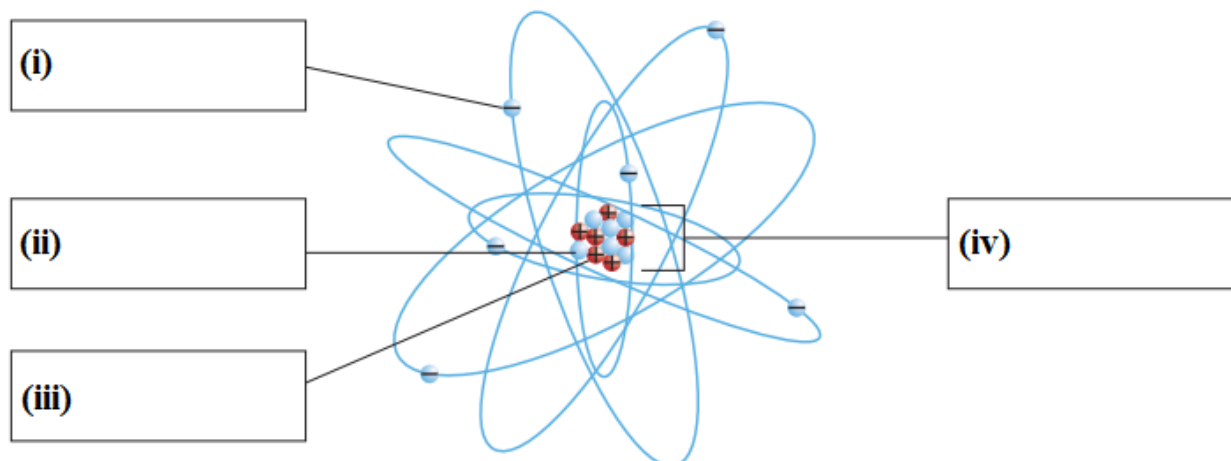
Atomic number	Mass number	No. of proton	No. of neutron	No. of electron	Name of element	Full atomic symbol
1	2					
2			2			
3	6					
4			5			
5	11					
6			8			
7	15					
8			9			
9	19					
10			11			
11	23					
12			13			
13	27					
14			15			
15	31					
16			18			
17	37					
18			22			
19	41					
20			24			

(20 marks)

(b) Which FOUR elements are most abundant in the Earth crust? Arrange them in decreasing order of abundance

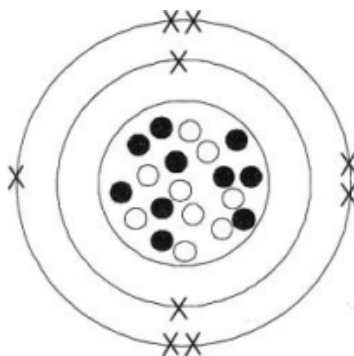
(1 mark)

2. The diagram below shows the simple model of an atom X.



- (a) Label the above diagram using these words: **proton, neutron, electron, nucleus.** (4 marks)
- (b) Write down the atomic number of atom X. (1 mark)
- (c) Write down the mass number of atom X. (1 mark)
- (d) With reference to the charges of the subatomic particles, explain why atom X is electrically neutral. (2 marks)
- (e) (i) With reference to the Periodic Table, write down the symbol of element that atom X belongs to. (1 mark)
- (ii) Hence, write down the name of the element that atom X belongs to. (1 mark)

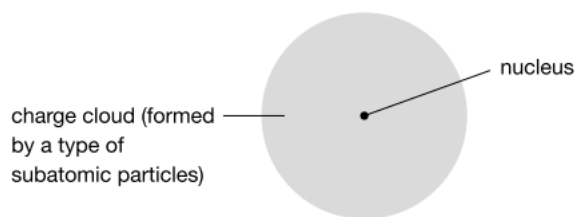
3. The following diagram shows the structure of an atom of element X. The crosses, black dots and white dots represent three different kinds of sub-atomic particles.



- (a) Identify the following sub-atomic particles:
- (i) Cross (1 mark)
 - (ii) Black dot (1 mark)
 - (iii) White dot (1 mark)
- (b) Give the full atomic symbol of the above atom (1 mark)
- (c) Explain why an atom is electrically neutral. (1 mark)
- (d) Tom claims that NOT all atoms in the world must contain protons, neutrons and electrons. Give one example to support his claim. (1 mark)

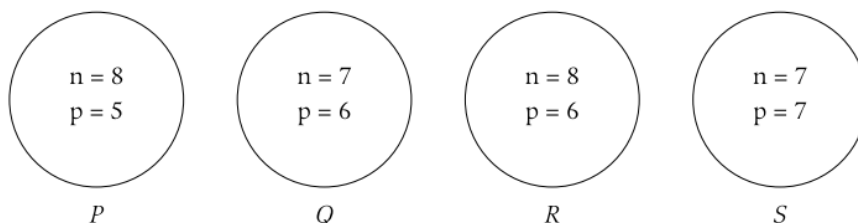
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4. The following simplified diagram represents the structure of the atom of a certain element with an atomic number greater than one.



- (a) Does the atom carry any net charges? Suggest why. (1 mark)
- (b) What type of subatomic particles can be found in the charge cloud? (1 mark)
- (c) Name TWO types of subatomic particles that can be found in the nucleus. (2 marks)
- (d) Explain why most of the mass of the atom is in the nucleus. (2 marks)

5. The diagram below shows the nuclei of four atoms P, Q, R and S.



n = number of neutrons;
 p = number of protons

- (a) State and explain which two atoms belong to the same element. (2 marks)
- (b) What is the element in (a)? (1 mark)
- (c) Write the full atomic symbol for each of these atoms. (4 marks)

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6. (a) Give the names and symbols of the following elements (A to D) from the descriptions given.

Element	Description	Name	Symbol
A	Its atomic number is 7		
B	Its atom has 12 protons		
C	Its atom has 18 electrons		
D	The electronic arrangement of its atom is 2,8,8,1		

(4 marks)

- (b) Draw electron diagrams of atoms of the following elements:

(i) Fluorine

(1 mark)

(ii) Calcium

(1 mark)

(iii) Neon

(1 mark)

(iv) Phosphorus

(1 mark)

7. Look at the table. It shows some information about the particles which make up atoms.

Particle	Charge	Relative Mass
Proton		1
Electron		0.005
Neutron	Neutral	

- (a) Complete the table.

(3 marks)

- (b) Look at the symbol for magnesium.



The mass number of magnesium is 25.

(i) What is meant by 'mass number'?

(1 mark)

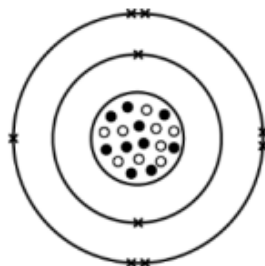
(ii) Write down the electronic arrangement of a magnesium atom.

(1 mark)

(iii) A magnesium atom is electrically neutral. Explain why.

(1 mark)

8. The diagram below represents the structure of an atom of element X. The crosses, black dots and white dots represent three different kinds of subatomic particles.



- (a) (i) What type of subatomic particle does the black dot represent? (1 mark)
- (ii) State the charge of the black dot. (1 mark)
- (b) Determine the atomic number and the mass number of X. (2 marks)
- (c) Hence, write the full atomic symbol of X. (1 mark)
9. Explain why the followings concerning atoms are INCORRECT.
- (a) All atoms contain protons, neutrons and electrons (1 mark)
- (b) There are about thousands of elements in the world (1 mark)
- (c) A proton and a neutron have exactly the same mass (1 mark)
- (d) The mass of an atom is evenly distributed within an atom (1 mark)
- (e) All elements have natural isotopes. (1 mark)

10. Deuterium (^2H) is an isotope of hydrogen. Deuterium reacts with chlorine-35 (^{35}Cl) to form a compound R.

(a) What is the mass number of deuterium?

(1 mark)

(b) Compound S formed from the reaction of chlorine-35 with another isotope of hydrogen protium (^1H). Do compounds R and S have the same chemical properties? Explain your answer.

(1 mark)

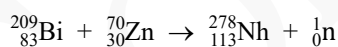
(c) Tritium is another isotope of hydrogen having 2 neutrons and 1 proton. Write the full atomic symbol for tritium.

(1 mark)

(d) Why does deuterium have an isotopic relationship with protium?

(1 mark)

11. Nihonium (Nh) is a metal element discovered by a team of Japanese scientists in 2004. It can be synthesized by the bombardment of bismuth with zinc.



(a) (i) Which types of subatomic particle does ^1_0n represent?

(1 mark)

(ii) Explain whether ^1_0n will be deflected by an electric field.

(1 mark)

(b) Calculate the number of neutrons in $^{278}_{113}\text{Nh}$.

(1 mark)

(c) Nihonium has several isotopes.

(i) What is meant by the term 'isotopes'?

(1 mark)

(ii) Predict, with examples, one similarity and one difference in properties among the isotopes of nihonium.

(2 marks)

12. Refer to the following information about fluorine and sulphur.

	Fluorine	Sulphur
Atomic number	9	16

- (a) (i) Draw the electron diagram of a fluorine atom. (1 mark)
- (ii) Draw the electron diagram of a sulphur atom. (1 mark)
- (b) All sulphur atoms have the same atomic number. But some sulphur atoms have different mass numbers. Explain briefly. (1 mark)
- (c) Fluorine-19 is the only isotope of fluorine. Describe the structure of a fluorine atom. Your answer should include the type and number of the subatomic particles present in the atom, and the location and the charge of each type of these subatomic particles. (3 marks)

13. There are 2 naturally occurring isotopes of element X, ^{35}X and ^{37}X . The abundance of ^{35}X is 75.78%.

- (a) Calculate the relative atomic mass of X. (2 marks)
- (b) What is the name of element X? (1 mark)
- (c) Write down the electronic arrangement of atom X. (1 mark)
- (d) State the number of neutrons in ^{37}X . (1 mark)

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14. Silver occurs in nature as two isotopes: ^{107}Ag and ^{109}Ag . The relative abundance of these isotopes is shown below.

Isotope	Relative abundance / %
^{107}Ag	5
^{109}Ag	45

- (a) State the meaning of the term 'relative atomic mass'.
(1 mark)
- (b) Calculate the relative atomic mass of silver.
(2 marks)
- (c) Is it possible to separate the isotopes of silver by chemical means? Explain your answer.
(2 marks)

15. Silicon (Si) has three naturally occurring isotopes.

- (a) What is the meaning of the term 'isotope'?
(1 mark)
- (b) The relative abundances of these isotopes are shown in the following table.

Isotope	$^{28}_{14}\text{Si}$	$^{29}_{14}\text{Si}$	$^{30}_{14}\text{Si}$
Relative abundance / %	92.23	4.68	?

- (i) How many protons and neutrons are there in $^{28}_{14}\text{Si}$?
(1 mark)
- (ii) What is the relative isotopic mass of $^{28}_{14}\text{Si}$?
(1 mark)
- (iii) What is the relative abundance of $^{30}_{14}\text{Si}$?
(1 mark)
- (iv) Calculate the relative atomic mass of silicon.
(2 marks)
- (v) Draw the electron diagram of an atom $^{28}_{14}\text{Si}$.
(1 mark)

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16. Element K has 3 naturally occurring isotopes, X, Y and Z. Consider the information below:

Isotope X: Isotopic mass of X is 29

Isotope Y: There are 16 neutrons in Y

Isotope Z: The number of protons, neutrons and electrons in Z are all 14

(a) Write down the full atomic symbol of X, Y and Z.

(3 marks)

(b) If the abundance of X is 5.6% and the relative atomic mass of K is 28.1, calculate the abundance of Y.

(2 marks)

17. Consider the information about element copper (Cu) below:

- Atomic number of copper is 29
- First three electron shells of copper are **fully filled**
- Copper has 4 occupied electron shells

(a) Write the electronic arrangement of a copper atom

(1 mark)

(b) Copper has TWO isotopes, ^{63}Cu and ^{65}Cu . Given that the relative atomic mass of copper is 63.5

(i) Calculate the abundance of ^{63}Cu .

(2 marks)

(ii) A sample of Cu metal contains 30624700 Cu atoms. Calculate the theoretical number of ^{65}Cu atom in the sample

(2 marks)

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18. (a) Element A occurs naturally in 2 isotopes with the abundance of each isotope shown in the table below:

Isotope	Abundance (%)
${}^6\text{A}$	7.59
${}^7\text{A}$	???

Calculate the relative atomic mass of element A.

(2 marks)

- (b) Element B occurs naturally in 3 isotopes with the abundance of each isotope shown in the table below:

Isotope	Abundance (%)
${}^{24}\text{B}$	79
${}^{25}\text{B}$	x
${}^{26}\text{B}$	y

The relative atomic mass of B is 24.32. Calculate the value of x.

(2 marks)

- (c) Element C occurs naturally in 3 isotopes with the abundance of each isotope shown in the table below:

Isotope	Abundance (%)
${}^{28}\text{C}$	92.2
${}^{29}\text{C}$	4.7
${}^n\text{C}$	3.1

The relative atomic mass of C is 28.11. Calculate the value of n.

(2 marks)

- (d) Element D occurs naturally in two isotopes, ${}^{63}\text{D}$ and ${}^{65}\text{D}$. If the abundance of ${}^{63}\text{D}$ is 75%, calculate the relative atomic mass of D.

(2 marks)

- (e) There is only one naturally occurring isotope of sodium, ${}^{23}\text{Na}$. Estimate the relative atomic mass of Na.

(1 mark)

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19. (a) Boron occurs naturally in two isotopes, ^{10}B and ^{11}B . Calculate the percentage abundance of ^{11}B .

(Relative atomic mass of B = 10.8)

(2 marks)

- (b) Silicon occurs naturally in 3 isotopes with the abundance of each isotope shown in the table below:

Isotope	Abundance (%)
^{28}Si	x
^{29}Si	7.8
^{30}Si	y

The relative atomic mass of Si is 28.1. Calculate the value of x.

(2 marks)

- (c) Boron occurs naturally in two isotopes, ^6Li and ^nLi . If percentage abundance of ^nLi is 90%, calculate the value of n.

(Relative atomic mass of Li = 6.9)

(2 marks)

- (d) Iridium occurs naturally in two isotopes, ^{191}Ir and ^{193}Ir . Calculate the theoretical number of ^{191}Ir in a sample of iridium containing 1.2×10^{21} iridium atoms. (Relative atomic mass of Ir = 192.25)

(3 marks)

- (e) Antimony has two naturally occurring isotopes, ^{123}Sb and ^aSb . The relative atomic mass of Sb is 121.8. A sample of antimony contains 4.3×10^{21} atoms. If the number of ^{123}Sb atoms in this sample is 1.84×10^{21} , calculate the value of a.

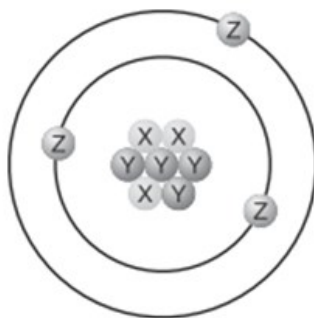
(3 marks)

20. Consider the table below:

Atom	Number of			Relative abundance
	Proton	Neutron	Electron	
^{25}Mg				11.0%
^{26}Mg				10.0%
^aMg		12		79.0%

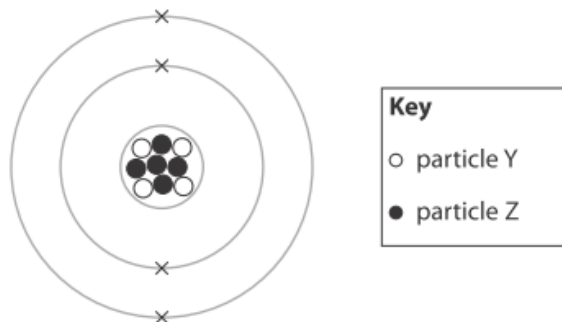
- (a) Complete the above table. (3 marks)
- (b) Calculate the value of a in the above table. (1 mark)
- (c) Calculate the relative atomic mass of Mg. (2 marks)
- (d) Draw the electron diagram of a Mg atom, showing electrons in ALL SHELLS. (1 mark)
- (e) When a sample of ^{25}Mg is heated strongly, a bright white flame can be observed. Predict what will happen if a sample of ^{26}Mg is heated strongly. Explain your answer. (1 mark)
- (f) State ONE different property of the three isotopes. (1 mark)

21. The diagram below represents the structure of an atom of isotope A of an element.



- (a) Name the following subatomic particles.
- (i) X (1 mark)
 - (ii) Y (1 mark)
 - (iii) Z (1 mark)
- (b) Write down the FULL ATOMIC SYMBOL of isotope A. (1 mark)
- (c) This element has two naturally occurring isotopes. In another isotope B, there are equal numbers of X, Y and Z. The ratio of abundance of A to B is 19 : 1
- (i) Write down the FULL ATOMIC SYMBOL of isotope B. (1 mark)
 - (ii) Calculate the relative atomic mass of this element. (Give your answer in 2 decimal places) (2 marks)
 - (iii) Explain why isotope A and isotope B have the same chemical properties. (1 mark)
- (d) In an atom of another element, there are 19 X and 20 Y.
- (i) What is the number of Z? (1 mark)
 - (ii) State the electronic arrangement of this element. (1 mark)
- (e) Tom claims that X, Y and Z are present in all atoms. Comment on his statement. (1 mark)
- (f) Do you expect isotope A carries net charge? Explain your answer. (1 mark)

22. The diagram shows the particles in the atom X of an element.



- (a) What are particle Y and particle Z respectively? (2 marks)
- (b) Write down the full atomic symbol of the atom X. (1 mark)
- (c) Atom W is an isotope of X. The mass number of W is 7.
- (i) What is meant by the term 'isotope'? (1 mark)
- (ii) State the numbers of particle Y and particle Z in atom W. (1 mark)
- (iii) Atom W is radioactive. In one of its decay process, a proton and an electron in atom W is combined to give a neutron to give a new atom V.
- (1) Write down the full atomic symbol of V. (1 mark)
- (2) Explain whether V and W have same chemical properties. (1 mark)

23. The atomic number of element Q is 14. It has three naturally occurring isotopes.

Isotope of Q	Number of neutrons	Relative abundance / %
Q_1	14	92.2%
Q_2	15	4.7%
Q_3	16	3.1%

(a) State the meaning for each of the following terms.

(i) Element

(1 mark)

(ii) Isotope

(1 mark)

(b) Refer to the above table, calculate the relative atomic mass of Q.

(2 marks)

(c) Draw the electron diagram for an atom of Q, showing electrons in the outermost shells only.

(1 mark)

24. The electronic arrangement of element Q is 2,8. It has three naturally occurring isotopes.

(a) State the meaning of the term 'isotopes'.

(1 mark)

(b) What is the name of element Q?

(1 mark)

(c) Refer to the following table, calculate the relative atomic mass of Q.

Isotope of Q	Number of neutrons	Relative abundance / %
Q_1	10	90.48
Q_2	11	0.27
Q_3	12	9.25

(2 marks)

Part 2

1. The following table shows some information about some elements. (A to F are NOT the symbols of the elements.)

Element	Melting point / °C	Boiling point / °C	Electrical conductivity
A	-7	58	poor
B	1083	2600	good
C	-39	357	good
D	-219	-183	poor
E	98	890	good
F	119	445	poor

- (a) Which of the above elements are liquids under room conditions? (1 mark)
- (b) State the temperature range in which F exists as a liquid. (1 mark)
- (c) (i) Which of the above elements are metals? (1 mark)
- (ii) State TWO properties of metals. (2 marks)
- (iii) A student stated that if a solid can conduct electricity, it must be a metal. Comment on the statement. (1 mark)
2. Element X has an atomic number of 15.
- (a) Draw the electron diagram for an atom of X. (1 mark)
- (b) In which period and group of the Periodic Table should X be placed? (2 marks)
- (c) By referring to the Periodic Table, name X. (1 mark)
- (d) Is X a metal or a non-metal? (1 mark)

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3. (a) If the atomic number of an element is known, its electronic arrangement, period number and group number can be written. Complete the following table without referring to the Periodic Table.

(P, Q, R, S, T and U are NOT the symbols of the elements.)

Element	Atomic number	Electronic arrangement	Period number	Group number
P	5			
Q	8			
R	10			
S	11			
T	14			
U	20			

(6 marks)

- (b) Which of the above elements is / are

(i) metals?

(1 mark)

(ii) non-metals?

(1 mark)

4. The table gives the melting and boiling points of four pure substances, W, X, Y and Z.

	Melting point / °C	Boiling point / °C
W	-7	60
X	660	2500
Y	180	1330
Z	115	445

Use data from the table to answer the questions

- (a) Which substance(s) is / are gas at 500 °C?

(1 mark)

- (b) Which substance is a liquid for the largest range of temperature?

(1 mark)

- (c) Which substance is a liquid at 1000 °C and a gas at 2000 °C?

(1 mark)

- (d) Substance W is an element which does not conduct electricity under room conditions. Suggest what W may be.

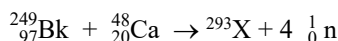
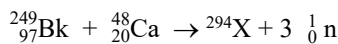
(1 mark)

5. The melting points and boiling points of four elements at 1 atmosphere are listed in the table below.

Elements	Melting point (°C)	Boiling point (°C)
Argon	-189	-186
Bromine	-7	59
Chlorine	-101	-35
Sulphur	119	445

- (a) Which of the above substances exist as a liquid at -90 °C and 1 atmosphere?
(1 mark)
- (b) The four substances are non-metals. Write FOUR typical properties, besides melting and boiling points, of non-metals.
(4 marks)
- (c) Draw electron diagrams for atoms of the following elements:
- (i) argon;
(1 mark)
- (ii) sulphur.
(1 mark)
6. Tin (Sn) is an element in Group IV of the Periodic Table. It has an atomic number of 50. Food cans made of iron are usually covered with a layer of tin for protection. The electronic arrangement of the tin atom is 2, 8, 18, r, s.
- (a) How many protons does a tin atom have?
(1 mark)
- (b) What are the values of r and s?
(2 marks)
- (c) In which period of the Periodic Table should tin be placed?
(1 mark)
- (d) Predict any TWO physical properties of tin.
(2 marks)

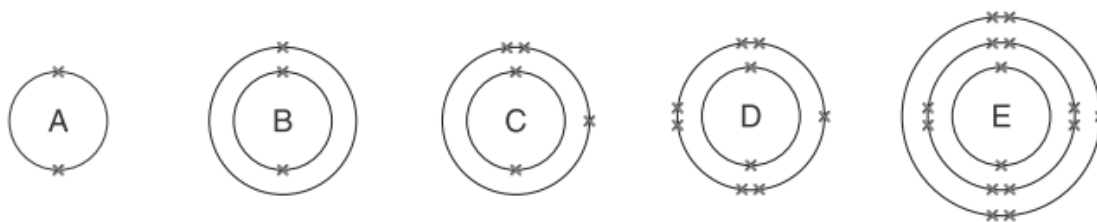
7. The nuclear fusion between berkelium (Bk) and calcium (Ca) atoms produces new atoms, either ^{293}X or ^{294}X , which have the same electronic arrangement 2, 8, 18, 32, 32, 18, 7. The reactions can be represented by the following equations:



- (a) What is the atomic number of element X? (1 mark)
- (b) What is the relationship between ^{293}X and ^{294}X ? (1 mark)
- (c) Calculate the number of neutrons in ^{294}X . (1 mark)
- (d) To which group of the Periodic Table does element X belong? Explain your answer. (2 marks)
8. (a) State the meaning of the term atomic number. (1 mark)
- (b) An atom of element X contains 14 protons, 14 electrons and 15 neutrons.
- (i) Which of these is the mass number of this atom? (1 mark)
- (ii) Explain which group of the Periodic Table element X belongs to. (1 mark)
- (c) The table shows the composition of a sample of a different element, Y, containing three isotopes.
- | Mass number of isotope | Percentage of isotope in sample |
|------------------------|---------------------------------|
| 32 | 95.0 |
| 33 | x |
| 34 | y |
- If the relative atomic mass of this sample of element Y is 32.0925, calculate the values of x and y respectively. (2 marks)

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9. This question is about atoms. Look at the diagrams. They show the atoms of some elements. The letters do NOT represent the symbols for the elements.



- (a) (i) Element B is in Group I. How can you tell?

(1 mark)

- (ii) Element E is in Period 3. How can you tell?

(1 mark)

- (b) The nucleus of an atom contains protons and neutrons.

- (i) Complete the table to show the relative charge and relative mass of a neutron.

	Relative charge	Relative mass
Proton	+1	1
Neutron		

(2 marks)

- (ii) There are two common types of chlorine atom, known as isotopes. One isotope is called chlorine-35 and the other is called chlorine-37. Complete the table below.

Symbol	Number of protons	Number of neutrons	Number of electrons
^{35}Cl			
^{37}Cl			

(2 marks)

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10. (a) Complete the following table:

Full Atomic symbol	Proton	Neutron	Electron	Mass number	Atomic number	Electronic arrangement	Name of element
	3	4					
	7			15			
	17			37			
		10	9				
			20	41			
			6	13			
		13			12		
				39	19		
				9	4		
		15					Silicon
				31			Phosphorus
				3			Helium
$^{39}_{18}\text{Ar}$		21					
$^{23}_{11}\text{Na}$				23			
$^{22}_{10}\text{Ne}$				22			
		62				2,8,18,18,1	
		119				2,8,18,32,18,2	
		116				2,8,18,32,17,1	
				133		2,8,18,18,8,1	
				204		2,8,18,32,18,4	
				197		2,8,18,32,18,1	

(b) Refer to the Periodic Table. Which of the above elements belong(s) to:

(i) Alkali metals?

(1 mark)

(ii) Alkaline earth metals?

(1 mark)

(iii) Halogens?

(1 mark)

(iv) Noble gases?

(1 mark)

		Group							
		I	II	III	IV	V	VI	VII	0
Period	2				a	b	c	d	e
	3		f		g				
	4	h							

12. Look at the diagram. It shows an outline of the periodic table.

[illegible]

Which symbol shows

- (a) an element whose atoms have 8 electrons in their outermost shell? (1 mark)
- (b) an element that is a red / brown liquid? (1 mark)
- (c) the element in Group 2 and Period 5? (1 mark)
- (d) an element that is a transition metal? (1 mark)
- (e) an element that is an alkali metal? (1 mark)

- (Note: the letters are NOT the symbols of the elements.)

	Group							
Period	I	II	III	IV	V	VI	VII	0
2	A					B		C
3		D		E			F	
4		G		H				

14. The diagram shows part of the Periodic Table, with elements represented by the letters L, M, Q, R and T. The letters in the diagram represent elements but are not their chemical symbols.

[illegible]

- (a) Give the letter from the diagram that represents a noble gas. (1 mark)
- (b) Elements L and M are in the same group. State why they have similar chemical reactions. (1 mark)
- (c) The atomic number of element Q is 31. Use this information to explain how you can determine the number of electrons in an atom of element R. (2 marks)

[illegible]

16. A to F are six different elements. Complete the following table according to the descriptions of elements

Element	Description	Period number	Group number	Electronic arrangement
A	Its atomic number is 13.			
B	Its name is helium			
C	Its atom has 9 electrons			
D	Its atom has 2 completely filled shells and 7 electrons in the 3 rd shell			
E	It is a non-metal but it can conduct electricity			
F	It is a period 2 metalloid			

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17. Consider the information given below:

Element	Description	Electronic arrangement of atom
A	<i>Most abundant element in air</i>	
B	There are 8 positively charged subatomic particles in the nucleus	
C	<i>A metal that burns with lilac flame under Bunsen flame</i>	
D	A period 3 element that does not have any chemical reaction with other substances	
E	A period 2 metalloid	

- (a) Complete the above table. (5 marks)
- (b) Draw the electron diagrams of element A, C and E. (3 marks)
- (c) What is the positively charged subatomic particles in the nucleus? (1 mark)
- (d) Explain why an atom is always electrically neutral (1 mark)
- (e) In terms of electronic arrangement, explain why element D does not react with other chemicals. (1 mark)
- (f) There are two natural isotopes of element E. Given that the abundance of ^{10}E is 20% and the relative atomic mass of element E is 10.8. Calculate the isotopic mass of another isotope. (2 marks)

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18. Rubidium (Rb) and potassium (K) are elements in Group I of the periodic table.

- (a) What is the name commonly given to this group of elements?
(1 mark)
- (b) The atomic number of rubidium is 37. The electronic arrangement of a rubidium atom is p,q,r,8,1. What are the values of p, q and r?
(1 mark)
- (c) Explain, in terms of electronic arrangements of their atoms, why rubidium and potassium have similar chemical properties.
(1 mark)
- (d) Predict, with ONE reason, whether rubidium or potassium is more reactive towards chlorine.
(1 mark)
- (e) A sample of rubidium consists of two isotopes, rubidium-85 and rubidium-87. Complete the following table about the atomic structures of these two isotopes.

Isotope	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons
Rubidium-85					
Rubidium-87					

(2 marks)

19. The Group I elements of the Periodic Table are shown below.

Li
Na
K
Rb
Cs
Fr

- (a) What is the name given to Group I elements?
(1 mark)
- (b) State one physical property of Group I elements.
(1 mark)
- (c) Potassium burns in air to form a white powder. What do you expect when rubidium burns in air? Explain your answer.
(2 marks)
- (d) Suggest how Group I metals are stored in the laboratory.
(1 mark)

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20. This question is about some elements in Group 1 of the Periodic Table.

(a) What is the special name of Group I element?

(1 mark)

(b) The table gives some statements about the reaction of potassium with water. Place ticks in three boxes to show which three statements are correct.

Statement	
potassium reacts more vigorously than sodium when added to water	
potassium sinks to the bottom of the water	
bubbles of oxygen gas are produced	
a lilac flame is seen	
potassium moves around	
a solution of potassium oxide is formed	

(1 mark)

(c) After the reaction of potassium with water is complete, a few drops of universal indicator are added to the solution formed. State, with explanation, what would be observed.

(2 marks)

21. The following table gives some information of the Group I metals.

Metal	Symbol	Atomic number	Electronic arrangement of atom
	Li	3	
	Na	11	
	K	19	
Rubidium	Rb	37	2, 8, 18, 8, 1

(a) Complete the above table.

(3 marks)

(b) What is the name given to the Group I metals?

(1 mark)

(c) Potassium has a much lower density than most metals. Give another property of potassium that is different from most other metals.

(1 mark)

(d) (i) State whether rubidium is more reactive than potassium.

(1 mark)

(ii) Suggest how rubidium can be stored safely in the laboratory.

(1 mark)

(iii) Suggest ONE safety precaution for handling rubidium in the laboratory.

(1 mark)

22. (a) How many outermost shell electrons do atoms of Group II elements have?
(1 mark)
- (b) The chemical properties of Group II elements are generally similar. Explain briefly.
(1 mark)
- (c) Does the reactivity of Group II elements increase or decrease down the group?
(1 mark)
- (d) Magnesium has almost no reaction with cold water, while calcium reacts with cold water at a moderate rate. Based on this information, predict the reactivity of beryllium (atomic number = 4) and barium (atomic number = 56) towards cold water.
(1 mark)
- (e) What is the gaseous product formed from the reaction between calcium and water? Suggest a test for this gas.
(2 marks)
23. Radium (Ra) is a Group II element. Its physical properties and chemical properties are similar to those of the other Group II elements.
- (a) How many electrons are there in the outermost shell in a radium atom?
(1 mark)
- (b) State any one similarity in physical properties among the Group II elements.
(1 mark)
- (c) When a small piece of radium is added to water, it fizzes vigorously and finally disappears to give a colourless solution.
- (i) Describe a test which can be used to show that the gas evolved is hydrogen.
(1 mark)
- (ii) State the colour of the solution if universal indicator is added to the resultant solution.
(1 mark)
- (d) Which of the following Group II elements is the least reactive?
strontium, magnesium, beryllium, radium, calcium, barium
(1 mark)

24. The diagram below shows part of elements in Group I and Group II in Periodic Table.

I	II
3 Li 6.9	4 Be 9.0
11 Na 23.0	12 Mg 24.3
19 K 39.1	20 Ca 40.1
37 Rb 85.5	38 Sr 87.6
55 Cs 132.9	56 Ba 137.3

- (a) Using magnesium as example, describe the appearance of Group I and Group II metals. (1 mark)
- (b) Which of the above metals is most reactive? (1 mark)
- (c) Give the special name of the following groups. (1 mark)
- (i) Group I (1 mark)
- (ii) Group II (1 mark)
- (d) Draw the electron diagram of Mg, showing electrons in ALL SHELLS. (1 mark)
- (e) Suggest an appropriate method to store group I metals. Explain your answer. (1 mark)
- (f) Both barium (Ba) and calcium (Ca) can react with water to give hydrogen gas. (1 mark)
- (i) Suggest a potential hazard when a piece of barium is added to water. (1 mark)
- (ii) Explain why Ba and Ca react with water similarly. (1 mark)
- (iii) Which metal reacts with water more vigorously? Explain your answer. (1 mark)
- (g) Suggest how sodium and potassium can be distinguished by the following methods. (1 mark)
- (i) Heating them strongly by non-luminous flame (1 mark)
- (ii) Measuring the melting point (1 mark)

25. Fluorine, chlorine, bromine and iodine belong to Group VII of the Periodic Table.

- (a) What is the name given to Group VII elements?
(1 mark)
- (b) State one similarity in the atomic structures of the Group VII elements.
(1 mark)
- (c) Which of the above elements is the most reactive?
(1 mark)
- (d) Fluorine and chlorine are gases under room conditions. Explain why we should NOT distinguish the two gases by smelling them directly.
(1 mark)

26. The table below gives some information about the first five elements in Group VII of the Periodic Table.

Element	Atomic number	Appearance	Melting point / °C	Boiling point / °C	Reaction with iron wool
Fluorine	9	Pale yellow gas	-220	-188	Reacts vigorously
Chlorine	17	Yellowish green gas	-101	-35	Reacts very rapidly with heated iron wool
Bromine	35	Reddish brown liquid	-7	59	Reacts moderately when the mixture is heated
Iodine	53	Black solid	114	184	Reacts slowly when the mixture is strongly heated
Astatine	85	?	?	?	?

- (a) Predict the appearance of astatine at room temperature and pressure. Explain your answer.
(3 marks)
- (b) Describe the variation in reactivity of halogens with atomic number.
(1 mark)
- (c) Iodine reacts with iron wool only when the mixture is strongly heated. Predict the reaction of astatine with iron wool when their mixture is strongly heated.
(1 mark)

27. This question is about three Group VII elements.

(a) Complete the following table.

Element	State at room temperature and pressure	Color
Chlorine		
Bromine		
Iodine		

(6 marks)

(b) What is the name given to Group VII of the periodic table?

(1 mark)

(c) Explain, in terms of electronic arrangements of their atoms, why the three elements have similar chemical properties.

(1 mark)

(d) Suggest whether the reactivity of Group VII elements increases or decreases down the group.

(1 mark)

(e) Draw an electron diagram of an atom of chlorine.

(1 mark)

28. Bromine is a reddish brown liquid that has a range of uses. It is one of the halogens.

(a) The atomic number of bromine is 35. The electronic arrangement of a bromine atom can be represented as 2,8,x,y.

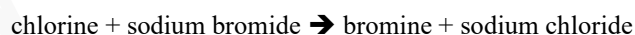
What are the values of x and y?

(2 marks)

(b) Suggest TWO gradual changes in the physical properties of halogens down the group.

(2 marks)

(c) Sea water contains sodium bromide. During the extraction of bromine from sea water, chlorine is bubbled through sea water. Chlorine displaces bromine from sodium bromide solution because it is a more reactive halogen.



(i) Name ONE other halogen that can displace bromine from sodium bromide. Explain your choice.

(2 marks)

(ii) Write a word equation for the displacement reaction suggested in (ii).

(1 mark)

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29. This question is about Group VII elements and their reactions.

(a) Fluorine has the smallest atoms in Group VII.

(i) What is the name of Group VII?

(1 mark)

(ii) State why fluorine has the smallest atoms in Group VII.

(1 mark)

(iii) State the numbers of occupied electron shells and the outermost shell electrons in an atom of iodine.

(2 marks)

(b) (i) The table gives descriptions of the reactions of some Group VII elements with iron wool. Complete the table by giving a description of the reaction of fluorine with iron wool.

Element	Description of reaction with iron wool
Fluorine	
Chlorine	- Does not need heating - Reacts quickly
Bromine	- Needs heating - Reacts slowly
Iodine	- Needs heating - Reacts very slowly

(2 marks)

(ii) State the relationship between the reactivity of the Group VII elements and the size of their atoms.

(1 mark)

30. Xenon is an element in the same group of helium. It is commonly used in flash lamps.
- (a) To which group of the Periodic Table does xenon belong? What is the name given to this group of elements?
(2 marks)
 - (b) How many outermost shell electrons are there in
 - (i) a helium atom?
(1 mark)
 - (ii) a xenon atom?
(1 mark)
 - (c) Predict the physical state of xenon under room conditions.
(1 mark)
 - (d) Would you expect xenon to be more or less reactive than helium? Explain your answer.
(1 mark)
31. Argon and neon are colourless gases. They are in Group 0 of the periodic table.
- (a) What is the name given to Group 0 of the periodic table?
(1 mark)
 - (b) Suggest why Group 0 elements are unreactive.
(1 mark)
 - (c) Argon is used to fill light bulbs. Why is argon, instead of air, used in electric light bulbs?
(1 mark)
 - (d) Suggest ONE use of neon.
(1 mark)

32. The diagram below shows part of elements in Group VII and Group 0 in Periodic Table.

VII	0
	2 He 4.0
9 F 19.0	10 Ne 20.2
17 Cl 35.5	18 Ar 40.0
35 Br 79.9	36 Kr 83.8
53 I 126.9	54 Xe 131.3

- (a) Give the special name of the following groups.
- (i) Group VII (1 mark)
- (ii) Group 0 (1 mark)
- (b) State the electronic arrangement of Kr. (1 mark)
- (c) Draw the electron diagram of Cl, showing electrons in ALL SHELLS. (1 mark)
- (d) Astatine (At) is another group VII element with atomic number of 85. Predict whether astatine or iodine is more reactive. Explain your answer. (1 mark)
- (e) A sample of Ar(g) and Xe(g) are cooled separately. Which one will be liquefied first? Explain your answer. (1 mark)
- (f) Using Ne as example, explain why Group 0 elements are very inert. (1 mark)
- (g) Aqueous bromine solution can be decolorized by sulphur dioxide gas. It is expected that aqueous iodine solution can also be decolorized by sulphur dioxide gas. Suggest a reason for this expectation based on electronic arrangement. (1 mark)

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6. This question is about chemical elements. Use the Periodic Table to help you answer this question.

- (a) (i) Identify the element with atomic number 5 (1 mark)
- (ii) Give the symbol of a metalloid in Period 3 (1 mark)
- (iii) Identify the element whose atoms contain 35 protons. (1 mark)
- (iv) Identify the element whose atoms have the electronic configuration 2,5. (1 mark)
- (b) The position of an element in the Periodic Table can be used to predict its properties.
- (i) Which group contains elements that are all unreactive? Explain your answer. (2 marks)
- (ii) Which element is the least reactive in Group I? Explain your answer. (2 marks)

34. A part of the periodic table is shown below:

		Group							
		I	II	III	IV	V	VI	VII	0
Period	2	Li	Be	B	C	N	O	F	Ne
	3	Na	Mg	Al	Si	P	S	Cl	Ar
	4	K	Ca					Br	Kr
	5								Xe

- (a) Across a period, the elements demonstrate a gradual change in some of their physical properties. State ONE such property. (1 mark)
- (b) In what way are the electronic arrangements of atoms of magnesium and calcium
- (i) different from each other? (1 mark)
- (ii) similar to each other? (1 mark)
- (c) Which metal and non-metal in the above table would react most vigorously with each other? (1 mark)
- (d) Sodium and magnesium are added separately to cold water in troughs. State TWO differences in the observations you expect. (2 marks)

[illegible]

36. A part of the Periodic Table is shown below, with some elements represented by letters T, U, V, X, Y and Z. (Note: the letters are NOT the symbols of the elements).

T																	
U																	V
X																	
Y										Z							

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37. A part of the Periodic Table is shown below.

		Group									
		I	II			III	IV	V	VI	VII	0
	1					H					He
	2	Li	Be			B	C	N	O	F	Ne
Period	3	Na	Mg			Al	Si	P	S	Cl	Ar
	4	K	Ca							Br	
	5									I	

- (a) Suggest why lithium and sodium are assigned to
- the same group. (1 mark)
 - different periods. (1 mark)
- (b) State TWO similarities in the physical properties of lithium and sodium. (2 marks)
- (c) Lithium reacts with water to give an alkaline solution and a colourless gas.
- Write the word equation for the reaction between lithium and water. (1 mark)
 - What is the product that makes the solution become alkaline? (1 mark)
- (d) Both magnesium and calcium are Group II elements. Predict, with one reason, whether magnesium or calcium is more reactive towards chlorine. (1 mark)
- (e) Which metal shown in the above Periodic Table reacts most vigorously with water? Suggest why. (1 mark)
- (f) Which element, fluorine or chlorine, is more reactive? Suggest why. (1 mark)
- (g) Suggest why Group 0 elements normally have little or no reaction with other elements. (1 mark)

38. Choose, from the 11 elements shown in the Periodic Table below, the ones that fit the following descriptions.

[illegible]

- (a) Three main group elements belong to the same period. State the Period number.
(2 marks)
- (b) Three elements belong to the same group. State the Group number.
(2 marks)
- (c) All the main group elements with two electrons in the outermost shell of their atoms.
(1 mark)
- (d) All elements that exist in gaseous state under room conditions.
(1 mark)
- (e) The metal that should be stored in paraffin oil.
(1 mark)
- (f) All the transition metals.
(1 mark)
- (g) The least reactive non-metal.
(1 mark)
- (h) All the elements with similar chemical properties as those of chlorine.
(1 mark)

39. A part of the periodic table is shown below:

	Group							
	I	II	III	IV	V	VI	VII	0
2	Li	Be	B	C	N	O	F	Ne
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca					Br	Kr

- (a) What determines the position of an element in the periodic table? (1 mark)
- (b) Across a period, the elements show a gradual change in some of their physical properties. State ONE such property. (1 mark)
- (c) For each of the following pairs of elements, suggest ONE reaction in which they behave similarly.
- (i) Sodium and potassium (2 marks)
- (ii) Chlorine and fluorine (1 mark)
- (d) Which metal and non-metal in the above table would react most vigorously with each other? (1 mark)
- (e) The atomic number of krypton (Kr) is 36. The electronic arrangement of a krypton atom is 2,8,x,y. What are the values of x and y? (1 mark)
- (f) Rubidium (Rb) is a Group I element. Its relative atomic mass is larger than that of potassium.
- (i) Suggest whether rubidium or potassium is more reactive. (1 mark)
- (ii) Suggest a method to store rubidium in the laboratory. (1 mark)

40. With reference to the part of the periodic table shown, answer the following questions.

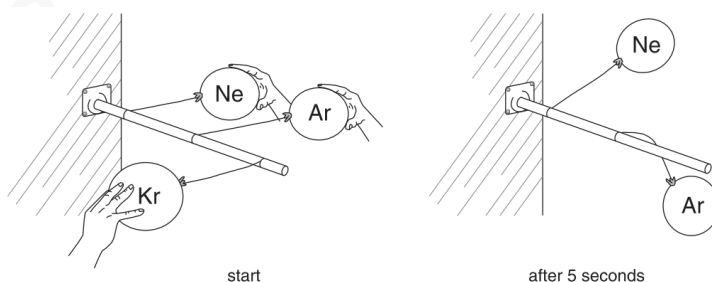
	Group							
	I	II	III	IV	V	VI	VII	0
2	Li	Be	B	C	N	O	F	Ne
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca					Br	

- (a) Name ONE metalloid in the above table. (1 mark)
- (b) Explain why argon is unreactive (1 mark)
- (c) Name the group to which lithium and sodium belong. (1 mark)
- (d) In what way are the electronic arrangements of atoms of lithium and sodium
- (i) similar to each other? (1 mark)
- (ii) different from each other? (1 mark)
- (e) Point out the gradual change in the reactivity of Group II elements down the group. (1 mark)
- (f) Suggest TWO gradual changes in the physical properties of Group VII elements down the group. (2 marks)

41. A portion of the periodic table is shown below.

	Group							
	I	II	III	IV	V	VI	VII	0
2	Li	Be	B	C	N	O	F	Ne
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca					Br	Kr
5							I	

- (a) How are the elements arranged in the periodic table? (1 mark)
- (b) What is the similarity in the electronic arrangement of atoms of elements in the same period? (1 mark)
- (c) Which element in Group I has the smallest atomic size? (1 mark)
- (d) Suggest why Group 0 elements seldom form compounds. (1 mark)
- (e) (i) Which is the most reactive metal in the above table? (1 mark)
- (ii) Suggest how the most reactive metal should be stored. (1 mark)
- (iii) State ONE hazard warning label that should be displayed on a reagent bottle containing that metal. (1 mark)
- (f) Which of the above elements is used to make computer chips? (1 mark)
- (g) Balloons filled with different noble gases were tied to a bar. They were held horizontally at the same height and then released. The positions of two of the balloons 5 seconds after release are shown in the diagram.



Predict the position of the balloon filled with krypton (Kr) after 5 seconds.

(1 mark)

Na												Al				S	Cl	
K																	Xe	
													In					

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43. The diagram below is a part of the Periodic Table:

	I	II	III	IV	V	VI	VII	0
2 nd	a	b			c	d	e	f
3 rd	g	h	i			j	k	l
4 th	m	n					o	p
5 th	q	r					s	

The letter **a** to **s** refer to some of the elements in the Periodic Table.

- (a) Suggest how elements are arranged in the Periodic Table. (1 mark)
- (b) Both **e** and **k** can be decolorized by sodium sulphite solution.
 - (i) Explain why **e** and **k** react with sodium sulphite solution similarly (1 mark)
 - (ii) Which element (**e** or **k**) reacts with sodium sulphite more vigorously? Explain your answer. (1 mark)
- (c) In terms of electronic arrangement, explain why element **l** does not form compound with other elements. (1 mark)
- (d) Which element in the above Table:
 - (i) is the most reactive non-metal? (1 mark)
 - (ii) is the most reactive metal? (1 mark)
- (e) Which of the above elements is the most abundant element in air? (1 mark)
- (f) Which of the above elements give brick-red flame in flame test? (1 mark)
- (g) Suggest a method to store element **q** in laboratory. Explain your answer. (1 mark)
- (h) Suggest a safety precaution when handling **k** in laboratory. Explain your answer. (1 mark)

44. The following diagram shows a part of the Periodic table.

I	II											III	IV	V	VI	VII
A													B	C		D
E	F											G	H			I
J	K			L					M						N	O
																P

Referring to the elements indicated in the above diagram only, answer the following questions by using the symbols stated in the above diagram. (A to P are NOT the element symbol)

- Which of the above elements is the most reactive metal? (1 mark)
- Which of the above elements is the most reactive non-metal? (1 mark)
- Write down the electronic arrangement of the following elements:
 - K (1 mark)
 - O (1 mark)
- Which of the above elements is / are transition metal? (1 mark)
- Which of the above elements exists as a liquid at room conditions? (1 mark)
- Which of the above halogen has the lowest boiling point? (1 mark)
- Which of the above elements is / are expected to share similar chemical properties with beryllium? Explain your answer. (2 marks)
- Identify ONE metalloid in the above elements. (1 mark)
- The isotopic mass of atom X is 23 and it has 12 neutrons. Which element does X belong to? (1 mark)
- There are 2 outermost shell electrons in an atom of M. Write down the electronic arrangement of atom M. (1 mark)

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45. Some of the sub-atomic information of atoms A to G is given in the table below.

Atom	Atomic number	Mass number	Proton	Neutron	Electron
A				21	20
B			10	11	
C	4			5	
D				18	17
E		40	20		
F		29			14
G	6	13			

- (a) Complete the above table (7 marks)
- (b) Give the special name of the group in the Periodic Table that atom A belongs to. (1 mark)
- (c) Give the special name of the group in the Periodic Table that atom B belongs to. (1 mark)
- (d) Give the FULL atomic symbols of atom A, E and G. (3 marks)
- (e) Which of the above elements CANNOT conduct electricity under room conditions? (1 mark)
- (f) Give one use of element of atom F. (1 mark)
- (g) Which of the above atoms has SAME chemical properties with atom A? Explain your answer. (1 mark)
- (h) Draw the electron diagram of atom A, atom B and atom C. (3 marks)

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46. Consider the information given below:

Element	Description	Name of element
A	An atom of A with isotopic mass of 39 has 20 neutrons	
B	There are 20 positively charged subatomic particles in the nucleus	
C	<i>An element that can be used to fill light bulb</i>	
D	Electronic arrangement of atom D is 2, 8, 2	
E	A noble gas with a duplet structure in its outermost shell	
G	<i>Most abundant metallic element in sea water</i>	
H	A period 3 metalloid	

- (a) Complete the above table. (8 marks)
- (b) Write down the electronic arrangement of an atom of element B (1 mark)
- (c) Element C can be used to fill light bulb as it is very chemically stable. In terms of electron arrangement, explain why element C is very chemically stable. (1 mark)
- (d) What is the flame color of element G? (1 mark)
- (e) Atom W is an atom of element H with 15 neutrons. Calculate the mass number of atom W. (1 mark)
- (f) Which of the above elements share(s) similar chemical properties with strontium, an alkaline earth metal with atomic number of 38? Explain your answer. (2 marks)
- (g) Suggest how to store element A in school laboratory. Explain your answer. (1 mark)

47. X, Y and Z are three different atoms. Refer to the descriptions on two atoms below.

Atom X: It has 12 neutrons and 11 electrons

Atom Y: It is a group I element with 4 occupied electron shells. The mass number of atom Y is 40

Atom Z: It has 3 occupied electron shells, 1 in the outermost shell. The mass number of atom Z is 23.

(a) A student claims that atom X and atom Z are isotopes

(i) What is meant by the term 'isotope'?

(1 mark)

(ii) Do you agree with his claim? Explain your answer.

(1 mark)

(iii) Explain why isotopes CANNOT be separated by chemical methods.

(1 mark)

(b) It is known that element of Y and element of Z can both react with water

(i) Predict which atom will react with water more vigorously.

(1 mark)

(ii) Explain why element of Y and element of Z have similar chemical properties.

(1 mark)

(iii) Name the group that atom Y belongs to in the Periodic Table.

(1 mark)

(c) Atom Y is radioactive. In the decay process of atom Y, one of its electrons and one of its protons will combine to form a neutron. Denote the new atom formed as atom W.

(i) Write a full atomic symbol of atom W.

(1 mark)

(ii) State the electronic arrangement of atom W.

(1 mark)

(iii) Give the special name of the group that atom W belongs to in the Periodic Table.

(1 mark)

(iv) Explain why atom W is electrically neutral.

(1 mark)

48. Lead (Pb) and carbon are elements that belong to the same group. The atomic number of lead is 82.

- (a) The electronic arrangement of lead is 2, 8, x, 32, x, y.
- (i) Calculate the values of x and y. (2 marks)
- (ii) State the group number of lead. Explain your answer. (2 marks)
- (iii) State the period number of lead. Explain your answer. (2 marks)
- (b) Lead atom can form a stable ion, lead(II) ion, by losing two of its outermost shell electrons.
- (i) State the number of protons and electrons in a lead(II) ion. (2 marks)
- (ii) What hazard warning label should be displayed on the reagent bottle containing lead(II) ion? (1 mark)
- (iii) Explain why the mass of a lead atom is almost the same as the mass of a lead(II) ion. (1 mark)
- (c) Scientists found that there are FOUR naturally occurring isotopes of lead.
- (i) What is meant by the term 'isotope'? (1 mark)
- (ii) Four naturally occurring isotopes of lead are ^{204}Pb , ^{206}Pb , ^{207}Pb and ^{208}Pb . Given that the abundance of ^{204}Pb , ^{207}Pb and ^{208}Pb are 1.4%, 22.1% and 52.4% respectively. Calculate the relative atomic mass of Pb. (2 marks)
- (iii) When ^{204}Pb is added to copper(II) sulphate solution, the solution turns from blue to colorless. Predict what will be observed if ^{208}Pb is added to copper(II) sulphate solution. Explain your answer. (1 mark)

49. Sodium is an example of metal while chlorine is an example of non-metal.
- (a) Explain why sodium and chlorine are placed in the same period in the Periodic Table (1 mark)
- (b) Lithium, sodium and potassium are placed in the same group in the Periodic Table. They are Group I elements. They are also known as Alkali Metals.
- (i) Explain why these metals are placed in the same group. (1 mark)
- (ii) Explain why they are described as 'alkali'. (1 mark)
- (iii) Arrange the above 3 elements in ascending order of reactivity. (1 mark)
- (iv) State one similarity in chemical property of Group I elements. (1 mark)
- (v) Suggest one safety precaution that should be taken when handling potassium. (1 mark)
- (c) Chlorine, bromine and iodine are also placed in the same group. They are Group VII elements.
- (i) Name the group that chlorine, bromine and iodine belong to. (1 mark)
- (ii) Arrange the above 3 elements in ascending order of reactivity (1 mark)
- (iii) State the change in color trend down the group of Group VII elements. (1 mark)
- (iv) State one similarity in chemical property of Group VII elements. (1 mark)
- (v) Suggest one safety precaution that should be taken when handling chlorine. Explain your answer. (2 marks)
- (vi) State the electronic arrangement of bromine (1 mark)

50. Compound X is formed when magnesium reacts with chlorine

(a) Explain why magnesium and chlorine are placed in the same period in the Periodic Table

(1 mark)

(b) Magnesium has 3 natural isotopes, ^{24}Mg , ^{25}Mg and ^{26}Mg . The natural abundance of ^{24}Mg and ^{25}Mg are 79% and 10% respectively.

(i) What is meant by the term 'isotope'?

(1 mark)

(i) State the number of protons, neutrons and electrons in a ^{25}Mg atom.

(1 mark)

(ii) Calculate the relative atomic mass of magnesium.

(2 marks)

(c) Chlorine has 2 natural isotopes, ^{35}Cl and ^{37}Cl . The relative atomic mass of chlorine is 35.5.

(i) State the number of protons, neutrons and electrons in a ^{37}Cl atom.

(1 mark)

(ii) Calculate the natural abundance of ^{37}Cl .

(2 marks)

(d) Magnesium also reacts with fluorine similarly. Predict whether magnesium reacts more vigorously with fluorine (compared with chlorine). Explain your answer.

(1 mark)

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51. Element X has 2 naturally occurring isotopes, ^{203}X and ^{205}X . Another element, element Y, also has 2 naturally occurring isotopes, ^{113}Y and ^{115}Y .
- (a) The electronic arrangement of X is 2, 8, 18, 32, 18, 3
- (i) State the group and period that X belongs to. (1 mark)
- (ii) Calculate the number of neutrons and protons in ^{203}X . (2 marks)
- (iii) Given that the relative atomic mass of X is 204.4, calculate the abundance of ^{203}X and ^{205}X . (2 marks)
- (b) Element Y has an electronic arrangement of 2, 8, 18, 18, 3. The abundance of ^{113}Y is 4.3%.
- (i) Calculate the relative atomic mass of Y. (2 marks)
- (ii) Explain whether a mixture containing X and Y can be separated by chemical methods. (1 mark)
- (iii) Explain whether a mixture containing ^{113}Y and ^{115}Y can be separated by chemical methods. (1 mark)
- (iv) Element Y can form two ions, Y^+ and Y^{3+} . Y^+ is formed when an atom of Y loses 1 outermost shell electron while Y^{3+} is formed when an atom of Y loses 3 outermost shell electrons.
- (1) Write down the electronic arrangement of Y^{3+} . (1 mark)
- (2) State the number of filled electron shells in a Y^+ ion. (1 mark)
- (c) It is known that both X and Y are metals.
- (i) Both X and Y can react with hydrochloric acid to give salt and hydrogen gas. Predict which element reacts with acid more vigorously. Explain your answer. (1 mark)
- (ii) The hydrochloric acid used in (i) contains chloride ions. Chloride ion is formed when chlorine atom gains electrons to attain the electronic arrangement of the nearest noble gas. State the electronic arrangement of a chloride ion. (1 mark)

52. The atomic number of lithium (Li) and rubidium (Rb) are 3 and 37 respectively. Lithium has 2 naturally occurring isotopes, ${}^6\text{Li}$ and ${}^7\text{Li}$. Rubidium also has 2 naturally occurring isotopes, ${}^{85}\text{Rb}$ and ${}^a\text{Rb}$ (a is the mass number of an isotope of Rb).
- (a) The relative atomic mass of Li is 6.9.
- (i) Explain why the relative atomic mass of Li is NOT an integer (1 mark)
- (ii) Calculate the abundance of ${}^6\text{Li}$. (2 marks)
- (b) The relative atomic mass of Rb is 85.5.
- (i) Given that the abundance of ${}^{85}\text{Rb}$ is 72.2%. Calculate the value of a in ${}^a\text{Rb}$. (2 marks)
- (ii) ${}^{85}\text{Rb}$ can react with oxygen in air to form an oxide. Predict whether ${}^a\text{Rb}$ would react with oxygen similarly. Explain your answer. (2 marks)
- (c) Write down the electronic arrangement of each of the following atoms:
- (i) Lithium atom (1 mark)
- (ii) Rubidium atom (Hint: Electronic arrangement of ${}_{36}\text{Kr} = 2,8,18,8$) (1 mark)
- (d) Lithium can react with water to give metal hydroxide and hydrogen gas.
- (i) With reference to your answer in (c), explain why rubidium also reacts with water to give metal hydroxide and hydrogen gas. (1 mark)
- (ii) Which metal reacts with water more vigorously? Explain your answer. (1 mark)

53. Iodine and bromine are Group VII elements that are placed under different periods. Under room conditions, iodine is a black solid while bromine is a brown liquid.
- (a) The atomic number of iodine is 53. Write down the electronic arrangement of an iodine atom, given that the first three electron shells of iodine are fully filled. (1 mark)
- (b) What is the special name given to Group VII? (1 mark)
- (c) There are two naturally occurring isotope of chlorine, ^{79}Br and ^{81}Br .
- (i) Given that the abundance of ^{79}Br and ^{81}Br is 11:9. Calculate the relative atomic mass of Br. (2 marks)
- (ii) State the number of neutrons in ^{81}Br . (1 mark)
- (iii) $^{79}\text{Br}_2$ can react with hydrogen to give steamy fumes HBr. State, with explanation, the expected observation when $^{81}\text{Br}_2$ reacts with hydrogen. (1 mark)
- (d) An aqueous solution of bromine (Br_2) is brown in color. It can be decolorized by adding sodium sulphite solution. It is expected that aqueous iodine (I_2) can also be decolorized by sodium sulphite solution. Suggest a reason for this expectation in terms of electronic arrangements. (1 mark)

54. Consider the table below:

Atom	Number of			Relative abundance
	Proton	Neutron	Electron	
^aSi		14		92.2%
^bSi		16		x
^{29}Si				y

- (a) Complete the missing parts of the above table. (3 marks)
- (b) Calculate the value of a and b in the above table. (2 marks)
- (c) The relative atomic mass of Si is 28.11.
- (i) What is meant by the term 'relative atomic mass'? (1 mark)
- (ii) Calculate the value of x in the above table. (1 mark)
- (d) State the electronic arrangement of a Si atom. (1 mark)
- (e) (i) Explain why silicon is classified as a semi-metal (1 mark)
- (ii) Suggest another semi-metal with atomic number less than 20. (1 mark)
- (f) Germanium (Ge) is also a semi-metal. Silicon and germanium are placed in the same group in the Periodic Table and the atomic number of Ge is 32.
- (i) Given that Ge is a Period 4 element. State the electronic arrangement of Ge. (1 mark)
- (i) Explain why Si and Ge have similar chemical properties. (1 mark)
- (g) Suggest one use of silicon in daily life. (1 mark)

PERIODIC TABLE 週期表

GROUP 族

atomic number 原子序

relative atomic mass 相對原子質量

		I		II												0					
3	Li	4	Be			21	22	23	24	25	26	27	28	29	30	5	6	7	8	9	10
6.9						Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	B	C	N	O	F	Ne
11	Na	12	Mg													13	14	15	16	17	18
23.0																Al	Si	P	S	Cl	Ar
24.3																27.0	28.1	31.0	32.1	35.5	40.0
19	K	20	Ca			39	40	41	42	43	44	45	46	47	48	31	32	33	34	35	36
						Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Ga	Ge	As	Se	Br	Kr
39.1										(98)	101.1	102.9	106.4	107.9	112.4	69.7	72.6	74.9	79.0	79.9	83.8
37	Rb	38	Sr			57 *										49	50	51	52	53	54
85.5						La										In	Sn	Sb	Te	I	Xe
55	Cs	56	Ba			138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	114.8	118.7	121.8	127.6	126.9	131.3
132.9						89 **										81	82	83	84	85	86
87	Fr	88	Ra			Ac	Rf	Db								Tl	Pb	Bi	Po	At	Rn
(223)						(227)	(261)	(262)								204.4	207.2	209.0	(209)	(210)	(222)
(226)																					

1

H

1.0

relative atomic mass 相對原子質量

		58	59	60	61	62	63	64	65	66	67	68	69	70	71
		Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
		140.1	140.9	144.2	(145)	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
		90	91	92	93	94	95	96	97	98	99	100	101	102	103
		Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
		232.0	(231)	238.0	(237)	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

*

**

Part 3

1. (a) Complete the following table:

Particle	Proton	Neutron	Electron	Mass number	Atomic number	Electronic arrangement
A	6	8	6			
B	13	14	10			
C	11	12	10			
D	10	12				2,8
E	35	46				2,8,18,8
F	82	125				2,8,18,32,18,2
G	7			15		2,8
H	17			37		2,8,7
I	19			41		2,8,8
J		5	2		4	
K		8	10		7	
L		16	18		15	
M				130	56	2,8,18,18,8,2
N				127	53	2,8,18,18,8
O				64	30	2,8,18,2
P			3	7	3	
Q			18	34	16	
R			20	41	20	

(18 marks)

(b) (i) Describe how an atom is changed to a cation.

(1 mark)

(ii) Which of the particles in the above table is / are cations?

(1 mark)

(c) (i) Describe how an atom is changed to an anion.

(1 mark)

(ii) Which of the particles in the above table is / are anions?

(1 mark)

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2. Consider an ${}^{27}_{13}\text{Al}$ atom
- (a) Write the numbers of protons, electrons and neutrons in the atom.
(3 marks)
 - (b) Draw the electron diagram of an ${}^{27}_{13}\text{Al}$ atom.
(1 mark)
 - (c) Describe briefly how an ${}^{27}_{13}\text{Al}$ atom can attain a stable octet electronic arrangement.
(1 mark)
 - (d) Write the numbers of protons, electrons and neutrons in the aluminium ion formed from an ${}^{27}_{13}\text{Al}$ atom.
(3 marks)
 - (e) Draw the electron diagram of an aluminium ion.
(1 mark)
3. In July 2009, the element with atomic number 112 was named copernicium. The element was named after Nicolaus Copernicus, a Polish astronomer. The symbol for an atom of copernicium is Cp.
- (a) The atomic number of copernicium is 112.
 - (i) How many protons are there in an atom of copernicium?
(1 mark)
 - (ii) Some atoms of copernicium have a mass number of 277. How many neutrons are in the nucleus of one of these atoms?
(1 mark)
 - (iii) By July 2009, four isotopes of copernicium were known. How do atoms of these isotopes of copernicium differ from each other?
(1 mark)
 - (b) As only a few atoms of copernicium have been made, its chemical and physical properties are unknown. However, copernicium is expected to form an ionic fluoride. This compound will contain copernicium ions, Cp^{2+} , and fluoride ions, F^{-} .
 - (i) Write the chemical formula of copernicium fluoride.
(1 mark)
 - (ii) Describe how copernicium atoms and fluorine atoms form copernicium ions and fluoride ions.
(2 marks)

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4. Ions can be formed when an atom gains or loses electrons.

(a) Complete the following table

Particle	Atomic number	Mass number	Proton	Neutron	Electron	Electronic arrangement
Li		6				
Li ⁺		7				
N				7		
N ³⁻				8		
K		39				
K ⁺		41				
Mg				13		
Mg ²⁺				14		
S		32				
S ²⁻		34				
F				9		
F ⁻				10		
Al		26				
Al ³⁺		27				
Ca				20		
Ca ²⁺				23		
Cl		35				
Cl ⁻		37				
Na				12		
Na ⁺				13		

(20 marks)

(b) Draw the electron diagram of the following particles.

(i) S²⁻ ion

(ii) Na atom

(iii) P³⁻ ion

(iv) Si atom

(v) Be²⁺ ion

(vi) Ar atom

(6 marks)

5. Complete the following table

Formula of particle	Atomic number	Mass number	Number of			Electron arrangement of particle
			Protons	Neutrons	Electrons	
^{13}C						
$^{23}\text{Na}^+$						
$^{16}\text{O}^{2-}$						
$^2\text{H}^-$						
$^{39}\text{K}^+$						
^{21}Ne						
	2	4			2	
	3	7			2	
	15	31			18	
	14			15	14	
	16			18	18	
	4			5	2	
		11	5		5	
		15	7		10	
		19	9		10	
			13	14	13	
			18	23	18	
			20	21	20	

6. (a) Complete the following table

Species	Atomic number	Mass number	Electronic arrangement	Number of		
				Protons	Neutrons	Electrons
(i) Beryllium atom	4				5	
(ii) Neon atom		20	2, 8			
(iii)				14	14	14
(iv) Phosphorus atom					16	
(v) Potassium ion	19	39				
(vi) Nitride ion			2, 8		7	
(vii)				12	12	10
(viii)		19	2, 8		10	
(ix)				11	12	10
(x)	11				12	11

(10 marks)

(b) Which species in the above table is / are negatively charged?

(1 mark)

(c) In which group of the periodic table should species (iii) be placed? Explain your answer.

(2 marks)

(d) In which period of the periodic table should species (vii) be placed? Explain your answer.

(2 marks)

(e) State the relationship between species (ix) and (x).

(1 mark)

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7. (a) Table 1 gives some information about three subatomic particles.

(i) Complete Table 1 by giving the missing information.

Subatomic particle	Relative mass	Relative charge
Electron	0.0005	
Proton		+1
Neutron	1	

(3 marks)

(ii) Give the name of the part of the atom containing protons and neutrons.

(1 mark)

(b) Table 2 shows the numbers of protons, neutrons and electrons in the species U, V, W, X, Y and Z.

Species	Number of protons	Number of neutrons	Number of electrons
U	8	10	8
V	9	10	10
W	11	12	10
X	11	12	11
Y	12	12	12
Z	12	13	12

Use the information in Table 2 to answer these questions. Each species may be used once, more than once or not at all.

(i) Give the letter of the species that has six electrons in its outer shell.

(1 mark)

(ii) Give the mass number of Z.

(1 mark)

(iii) Give the letter of the species that is a positive ion.

(1 mark)

(iv) Give the letters of the two species that are isotopes of the same element.

(1 mark)

(c) A sample of neon contains two isotopes, ^{20}Ne and ^{22}Ne .

(i) State ONE difference between ^{20}Ne and ^{22}Ne

(1 mark)

(ii) If the relative atomic mass of this sample of neon is 20.176, calculate the abundance of ^{20}Ne .

(2 marks)

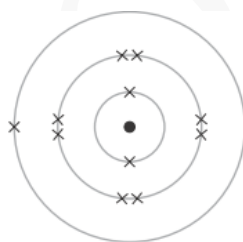
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8. The table shows the number of protons, neutrons and electrons in species F, G and H.

	F	G	H
Number of protons	7	7	7
Number of neutrons	7	8	7
Number of electrons	7	7	10

- (a) Give the mass number of F. (1 mark)
- (b) Give the electronic configuration of G. (1 mark)
- (c) State the relationship between F and G. Explain your answer. (1 mark)
- (d) State the relationship between F and H. Explain your answer. (1 mark)

9. The diagram shows the electronic configuration of an atom of an element.



- (a) Name the part of the atom that contains the protons and neutrons. (1 mark)
- (b) Give the number of protons in this atom. (1 mark)
- (c) Give the special name of the group this element belongs to. (1 mark)
- (d) Give the number of the period that contains this element. (1 mark)
- (e) Give the charge on the ion formed from this atom. (1 mark)

10. (a) When sodium reacts with chlorine, sodium chloride is formed.
- (i) (1) In terms of electron transfer, describe how sodium ion is formed from sodium atom. (1 mark)
 - (2) Write down the chemical formula of sodium ion, (1 mark)
 - (3) Write down the electronic arrangement of sodium ion. (1 mark)
- (ii) (1) In terms of electron transfer, describe how chloride ion is formed from chlorine atom. (1 mark)
- (2) Write down the chemical formula of chloride ion. (1 mark)
 - (3) Write down the electronic arrangement of chloride ion. (1 mark)
- (iii) With reference to your answer in (a)(i)(2) and (a)(ii)(2), write down the chemical formula of sodium chloride. (1 mark)
- (iv) Draw the electron diagram of sodium chloride, showing electrons in the outermost shells only. (1 mark)
- (b) When potassium reacts with nitrogen, potassium nitride is formed.
- (i) (1) In terms of electron transfer, describe how potassium ion is formed from potassium atom. (1 mark)
 - (2) Write down the chemical formula of potassium ion. (1 mark)
 - (3) Write down the electronic arrangement of potassium ion. (1 mark)
- (ii) (1) In terms of electron transfer, describe how nitride ion is formed from nitrogen atom. (1 mark)
- (2) Write down the chemical formula of nitride ion. (1 mark)
 - (3) Write down the electronic arrangement of nitride ion. (1 mark)
- (iii) With reference to your answer in (b)(i)(2) and (b)(ii)(2), write down the chemical formula of potassium nitride. (1 mark)
- (iv) Draw the electron diagram of potassium nitride, showing electrons in the outermost shells only. (1 mark)

- (c) When aluminium reacts with sulphur, aluminium sulphide is formed.
- (i) (1) In terms of electron transfer, describe how aluminium ion is formed from aluminium atom. (1 mark)
- (2) Write down the chemical formula of aluminium ion (1 mark)
- (3) Write down the electronic arrangement of aluminium ion. (1 mark)
- (ii) (1) In terms of electron transfer, describe how sulphide ion is formed from sulphur atom. (1 mark)
- (2) Write down the chemical formula of sulphide ion. (1 mark)
- (3) Write down the electronic arrangement of sulphide ion. (1 mark)
- (iii) With reference to your answer in (c)(i)(2) and (c)(ii)(2), write down the chemical formula of aluminium sulphide. (1 mark)
- (iv) Draw the electron diagram of aluminium sulphide, showing electrons in the outermost shells only. (1 mark)
- (d) When calcium reacts with oxygen, calcium oxide is formed.
- (i) (1) In terms of electron transfer, describe how calcium ion is formed from calcium atom. (1 mark)
- (2) Write down the chemical formula of calcium ion (1 mark)
- (3) Write down the electronic arrangement of calcium ion. (1 mark)
- (ii) (1) In terms of electron transfer, describe how oxide ion is formed from oxygen atom. (1 mark)
- (2) Write down the chemical formula of oxide ion. (1 mark)
- (3) Write down the electronic arrangement of oxide ion. (1 mark)
- (iii) With reference to your answer in (d)(i)(2) and (d)(ii)(2), write down the chemical formula of calcium oxide. (1 mark)
- (iv) Draw the electron diagram of calcium oxide, showing electrons in the outermost shells only. (1 mark)

11. Metals can react with non-metals to give ionic compounds.
- (a) What is ionic bond?
(1 mark)
- (b) When magnesium reacts with chlorine, magnesium chloride is formed.
- (i) State the electron arrangement of the following ions.
- (1) Magnesium ion
(1 mark)
- (2) Chloride ion
(1 mark)
- (ii) Write down the chemical formula of magnesium chloride.
(1 mark)
- (iii) Draw the electron diagram of magnesium chloride, showing electrons in the outermost shells only.
(1 mark)
- (c) When aluminium reacts with oxygen, aluminium oxide is formed.
- (i) State the electron arrangement of the following ions.
- (1) Aluminium ion
(1 mark)
- (2) Oxide ion
(1 mark)
- (ii) Write down the chemical formula of aluminium oxide.
(1 mark)
- (iii) Draw the electron diagram of aluminium oxide, showing electrons in the outermost shells only.
(1 mark)
- (d) When potassium reacts with hydrogen, potassium hydride is formed.
- (i) State the electron arrangement of the following ions.
- (1) Potassium ion
(1 mark)
- (2) Hydride ion
(1 mark)
- (ii) Write down the chemical formula of potassium hydride.
(1 mark)
- (iii) Draw the electron diagram of potassium hydride, showing electrons in the outermost shells only.
(1 mark)

12. Strontium chloride is commonly used as an active ingredient in toothpaste for sensitive teeth. Strontium chloride deposits in the pores (small holes) in dentine and forms a barrier to nerve endings of the teeth. Thus, it helps reduce the sensitivity of the teeth to temperature changes.

- (a) Describe the formation of strontium chloride from atoms of strontium and chlorine, and state the type of chemical bond involved.

(2 marks)

- (b) Draw the electron diagram of strontium chloride, showing electrons in the outermost shells only.

(1 mark)

- (c) Write the formula for strontium chloride.

(1 mark)

13. Consider the four elements W, X, Y and Z. (Note: the letters are NOT the symbols of the elements.)

Element	W	X	Y	Z
Atomic number	6	8	10	12

- (a) In which of these elements can its atoms form cations?

(1 mark)

- (b) State how a cation forms from an atom of the element mentioned in (a).

(1 mark)

- (c) Two of these elements react to form an ionic compound. Give the formula of the compound formed.

(1 mark)

- (d) Draw the electron diagram of the ionic compound formed in (c) (showing electrons in the outermost shells only).

(1 mark)

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14. Draw the electron diagrams, showing electrons in the outermost shell only, of the compounds formed from the following pairs of elements.

(a) Sodium and Chlorine	(b) Potassium and Bromine
(c) Fluorine and Lithium	(d) Calcium and Chlorine
(e) Oxygen and Sodium	(f) Magnesium and Fluorine
(g) Aluminium and Oxygen	(h) Calcium and Nitrogen
(i) Aluminium and Fluorine	(j) Magnesium and Oxygen

(10 marks)

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15. The following table shows some information about the atoms of four elements.

Element	Number of protons in atom	Number of neutrons in atom
W	3	4
X	7	7
Y	16	16
Z	19	20

- (a) Which of the above elements belong to the same group? Explain your answer. (1 mark)
- (b) W and X combine to form an ionic compound P.
- (i) Draw an electron diagram of P, showing electrons in the outermost shells only. (1 mark)
- (ii) State the chemical formula of P. (1 mark)
- (c) Y and Z combine to form an ionic compound Q.
- (i) Draw an electron diagram of Q, showing electrons in the outermost shells only. (1 mark)
- (ii) State the chemical formula of Q. (1 mark)

16. Element W belongs to Group II of the Periodic Table. It reacts with element Y to give an ionic compound with the formula of W_3Y_2 .

- (a) What is the charge on the ion of Y? (1 mark)
- (b) To which group does Y belong? (1 mark)
- (c) Draw the electron diagram for W_3Y_2 , showing electrons in the outermost shells only. (1 mark)
- (d) Write the formula of the compound formed between W and sulphur. (1 mark)

17. V, W, X, Y and Z are elements in the Periodic Table.
- W, X and Y are all Alkaline Earth Metal. Sum of their atomic number = 36
 - Ion of W has a duplet structure in its outermost shell
 - V and W are placed in the same period in the Periodic Table. Sum of their atomic number = 11
 - Ion of V and ion of Y have same electronic arrangement
 - X and Z are placed in the same period in the Periodic Table. Sum of their atomic number = 55
- (a) State the electronic arrangement of V, W, X, Y and Z (5 marks)
- (b) Draw the electron diagram, showing electrons in the outermost shell only, of the compounds formed between following pairs of atoms
- (i) V and X (1 mark)
 - (ii) X and Z (1 mark)
 - (iii) Aluminium and Z (1 mark)
 - (iv) Sodium and V (1 mark)
18. W, X and Y are three different elements. The ions formed from W and X, W^{2+} and X^{3-} , have the same electronic arrangement as a neon atom. (Note: W, X and Y are NOT symbols of elements.)
- (a) Suggest what W and X are. (2 marks)
- (b) Draw the electron diagram for the compound formed between W and X, showing electrons in the outermost shells only. (1 mark)
- (c) Y belongs to the same period as W and the same group as X in the Periodic Table. Suggest what Y is. (1 mark)

19. This question is about lithium oxide.

(a) Draw the electron diagram for each of the following atoms, showing electrons in ALL SHELLS

(i) Lithium atom

(1 mark)

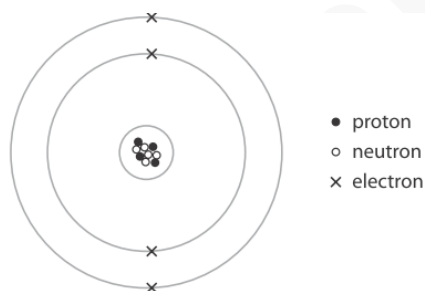
(ii) Oxygen atom

(1 mark)

(b) Describe the changes in electronic configuration when lithium and oxygen react to form lithium oxide.

(3 marks)

20. The diagram represents an atom of element X.



(a) Complete the following table:

Atomic number of this atom	
Mass number of this atom	
Period number of this element	
Number of electrons in the stable ion formed from this atom	

(4 marks)

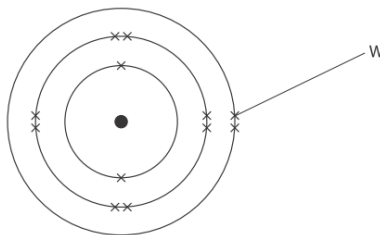
(b) In terms of sub-atomic particles, state a similarity and a difference for isotopes of the same element.

(2 marks)

(c) Under high temperatures, element X can react with nitrogen to form compound Y. Draw the electron diagram of compound Y, showing electrons in the outermost shells only.

(1 mark)

21. The diagram represents an atom of an element.



- (a) What is W? (1 mark)
- (b) An atom of this element has a mass number of 25. State the number of protons, neutrons and electrons in this atom. (1 mark)
- (c) State why atoms have no overall charge (1 mark)
- (d) What is the charge on the ion usually formed from this element? (1 mark)
- (e) This element can form a compound with iodine. Draw the electron diagram of the compound formed, showing electrons in the outermost shells only. (1 mark)

22. Strontium (Sr) and calcium (Ca) are elements in Group II of the periodic table.

- (a) What is the name commonly given to this group of elements? (1 mark)
- (b) State the similarity between a calcium atom and a strontium atom in terms of electronic arrangement. (1 mark)
- (c) Strontium reacts with nitrogen to form an ionic compound. Draw an electron diagram of the compound, showing electrons in the outermost shells only. (1 mark)
- (d) A sample of strontium contains three isotopes: ^{86}Sr , ^{87}Sr and ^{88}Sr . The relative atomic mass of this sample is 87.73. The relative abundance of the ^{88}Sr isotope is 83.1%. What is the relative abundance of the ^{86}Sr ? (2 marks)

23. The following table shows some information about some species. They can be atoms or ions.

	Atomic number	Mass number	Number of		
			Protons	Electrons	Neutrons
A			8	8	8
B	8	18		8	
C			8	10	8
D	13			10	13
E			12	10	12
F	12	24		12	

- (a) Complete the above table. (6 marks)
- (b) What is the relationship between A and B? (1 mark)
- (c) What is the relationship between A and C? (1 mark)
- (d) What is the relationship between E and F? (1 mark)
- (e) X is a compound formed from A and D, while Y is a compound formed from B and E.
- (i) Draw an electron diagram of X. (1 mark)
- (ii) Draw an electron diagram of Y. (1 mark)

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24. Consider the following information about 7 atoms:

Atom	A	B	C	D	E	F	G	H
Atomic number	12	9	3	20	14	16	7	18

- (a) Which of the above elements is / are:
- (i) Metal? (1 mark)
 - (ii) Non-metal? (1 mark)
 - (iii) Metalloid? (1 mark)
- (b) Give the special name of the group of the following elements belongs to in the Periodic Table.
- (i) Atom A (1 mark)
 - (ii) Atom B (1 mark)
 - (iii) Atom C (1 mark)
 - (iv) Atom H (1 mark)
- (c) Draw the electron diagram, showing electrons in the outermost shell only, of the compounds formed between following pairs of atoms
- (i) A and F (1 mark)
 - (ii) A and G (1 mark)
 - (iii) B and D (1 mark)
 - (iv) C and G (1 mark)

25. When potassium burns in oxygen, potassium oxide is produced.
- (a) With the help of a simple diagram, describe the bonding in potassium metal.
(2 marks)
 - (b) Describe the formation of ionic bond in potassium oxide.
(2 marks)
 - (c) Draw the electron diagram for potassium oxide, showing electrons in the outermost shells only.
(1 mark)
26. This question compares the bonding of sodium and sodium oxide.
- (a) Sodium, Na, is a metallic element. Explain, with the aid of a labelled diagram, what is meant by the term 'metallic bonding'.
(2 marks)
 - (b) Sodium reacts with oxygen to form sodium oxide, Na_2O , which is an ionic compound.
 - (i) State what is meant by the term 'ionic bond'.
(1 mark)
 - (ii) Draw an electron diagram to show the bonding in Na_2O , showing electrons in the outermost shells only.
(1 mark)

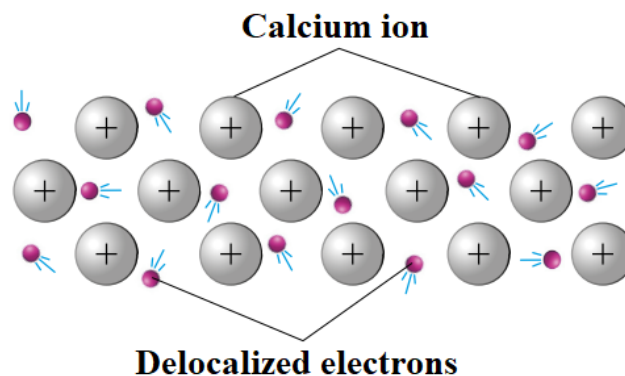
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27. Magnesium occurs naturally in three isotopes with the abundance of each isotope shown in the table below:

Isotope	Abundance (%)
^{24}Mg	79
^{25}Mg	x
^{26}Mg	y

- (a) What is meant by the term 'isotope'? (1 mark)
- (b) The relative atomic mass of Mg is 24.32. Calculate x. (2 marks)
- (c) Explain why magnesium is placed in Group II in the Periodic Table. (1 mark)
- (d) Magnesium oxide is an oxide of magnesium. It can be obtained by heating magnesium in air. Draw the electron diagram of magnesium oxide, showing electrons in the outermost shells only. (1 mark)
- (e) With the aid of a labelled diagram, describe the bonding of magnesium (3 marks)
- (f) Radium is also an alkaline earth metal. The electronic arrangement of an atom of radium is 2, 8, 18, 32, 18, 8, 2.
- (i) State the atomic number of radium. (1 mark)
- (ii) Predict whether Mg or element A has a higher reactivity towards chlorine. Explain your answer. (1 mark)

28. The diagram below shows the structure of calcium metal.



- (a) Draw the electron diagram of a calcium ion, showing electrons in ALL SHELLS. (1 mark)
- (b) How many delocalized electrons can be contributed by ONE calcium atom? (1 mark)
- (c) With reference to the above graph, briefly describe the formation of metallic bond. (1 mark)
- (d) Explain why solid calcium is a good conductor of electricity. (1 mark)
- (e) Calcium is a reactive metal. It can react with oxygen in air readily to form an ionic compound X.
 - (i) Briefly describe the formation of ionic bond when calcium reacts with oxygen to give X. (1 mark)
 - (ii) Draw the electron diagram of X, showing electrons in the outermost shells only. (1 mark)
 - (iii) At higher temperatures, calcium also reacts with nitrogen in air to give compound Y. Draw the electron diagram of Y, showing electrons in the outermost shells only. (1 mark)

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29. Complete the following table

	Chemical name	Chemical formula
(a)	Sodium chloride	
(b)	Calcium chloride	
(c)	Copper(II) chloride	
(d)	Potassium oxide	
(e)	Sodium nitrate	
(f)	Iron(II) sulphate	
(g)	Potassium dichromate	
(h)	Silver sulphate	
(i)	Ammonium dichromate	
(j)	Zinc nitrate	
(k)	Aluminium sulphate	
(l)	Chromium(III) sulphate	
(m)	Calcium sulphite	
(n)	Lead(II) sulphate	
(o)	Sodium hydroxide	
(p)	Iron(III) hydroxide	

30. Complete the following table

	Chemical name	Chemical formula
(a)		KI
(b)		AlBr ₃
(c)		Na ₃ N
(d)		FeCl ₃
(e)		<u>FeS</u>
(f)		AgCl
(g)		NaHCO ₃
(h)		CaCO ₃
(i)		(NH ₄) ₂ CO ₃
(j)		KNO ₂
(k)		Al(OH) ₃
(l)		FeSO ₄
(m)		Fe ₂ (SO ₄) ₃
(n)		Pb(NO ₃) ₂
(o)		Cr(OH) ₃
(p)		CuCO ₃

31. Give chemical formulae to the following ionic compounds

	Chemical name	Chemical formulae
(a)	Silver nitrate	
(b)	Sodium hydroxide	
(c)	Potassium sulphate	
(d)	Calcium oxide	
(e)	Sodium sulphite	
(f)	Calcium hydroxide	
(g)	Iron(III) nitrate	
(h)	Magnesium sulphide	
(i)	Iron(II) chloride	
(j)	Iron(III) oxide	
(k)	Calcium nitride	
(l)	Lead(II) nitrate	
(m)	Zinc hydroxide	
(n)	Aluminium iodide	
(o)	Sodium nitrite	
(p)	Copper(I) oxide	
(q)	Aluminium sulphate	
(r)	Sodium phosphide	
(s)	Potassium dichromate	
(t)	Magnesium carbonate	
(u)	Chromium(III) oxide	
(v)	Calcium hydrogencarbonate	
(w)	Iron(III) phosphate	
(x)	Ammonium dichromate	
(y)	Potassium permanganate	
(z)	Iron(III) sulphate	

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32. Give full names of the following ionic compounds

	Chemical formulae	Chemical name
(a)	FeCl_2	
(b)	CuSO_4	
(c)	Na_2S	
(d)	AgBr	
(e)	CaF_2	
(f)	K_2SO_3	
(g)	CaSO_4	
(h)	$(\text{NH}_4)_2\text{SO}_4$	
(i)	AlCl_3	
(j)	$\text{Fe}(\text{OH})_2$	
(k)	ZnCO_3	
(l)	MgCO_3	
(m)	Na_3PO_4	
(n)	AlN	
(o)	$\text{K}_2\text{Cr}_2\text{O}_7$	
(p)	CaSO_3	
(q)	KNO_2	
(r)	CuCr_2O_7	
(s)	Na_3P	
(t)	Fe_2O_3	
(u)	FePO_4	
(v)	$\text{Al}_2(\text{SO}_4)_3$	
(w)	$\text{Zn}(\text{NO}_3)_2$	
(x)	$(\text{NH}_4)_2\text{Cr}_2\text{O}_7$	
(y)	PbCl_2	
(z)	NaHCO_3	

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33. Complete the table below by filling in the name and colour of the aqueous solution of each compound.

Formula of compound	Name of compound	Color of its aqueous solution
NaMnO_4		
FeCl_2		
NiSO_4		
$\text{Fe}(\text{NO}_3)_3$		
CoCl_2		
NH_4Cl		
KNO_2		
K_2CrO_4		
CuBr_2		
$\text{K}_2\text{Cr}_2\text{O}_7$		

(10 marks)

34. Complete the following table.

Chemical name	Chemical formula	Color of aqueous solution
Potassium dichromate		
	ZnCl_2	
Sodium chromate		
	$\text{Cu}(\text{NO}_3)_2$	
Ammonium carbonate		
	KNO_2	
Nickel(II) chloride		
	FeSO_4	
Calcium hydroxide		
	$\text{Fe}_2(\text{SO}_4)_3$	
Sodium phosphate		
	KMnO_4	
Chromium(III) sulphate		
	AgNO_3	
Cobalt(II) bromide		
	NaHSO_4	
Aluminum sulphate		
	Na_2SO_3	
Calcium hydrogencarbonate		
	MnSO_4	

(20 marks)

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35. Draw the electron diagram of the following compounds, showing electrons in the outermost shell only.

(a) Calcium oxide	(b) Aluminium chloride
(c) Sodium sulphide	(d) Potassium nitride
(e) Magnesium fluoride	(f) Lithium oxide
(g) Magnesium phosphide	(h) Aluminium sulphide
(i) Calcium iodide	(j) Potassium phosphide

(10 marks)

36. The table gives the formulae of some positive and negative ions

	Na^+	Mg^{2+}
Cl^-	Compound A	Compound B
O^{2-}	Compound C	Compound D
N^{3-}	Compound E	Compound F

(a) Write down the chemical formula for each of the following compounds.

(i) Compound A

(1 mark)

(ii) Compound B

(1 mark)

(iii) Compound C

(1 mark)

(b) Draw the electron diagram for each of the following compounds, showing electrons in the outermost shells only.

(i) Compound D

(1 mark)

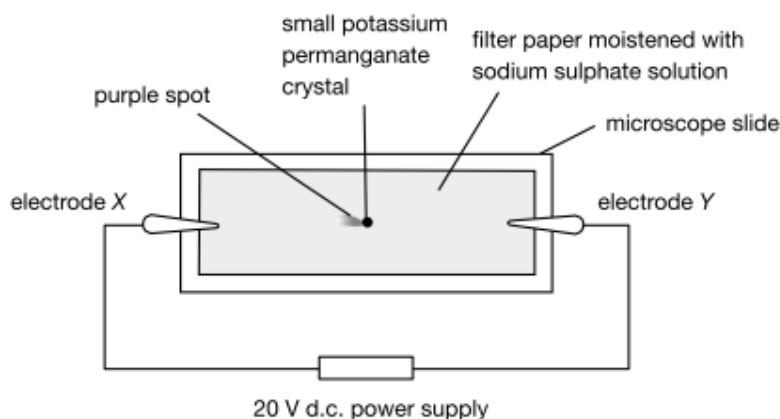
(ii) Compound E

(1 mark)

(iii) Compound F

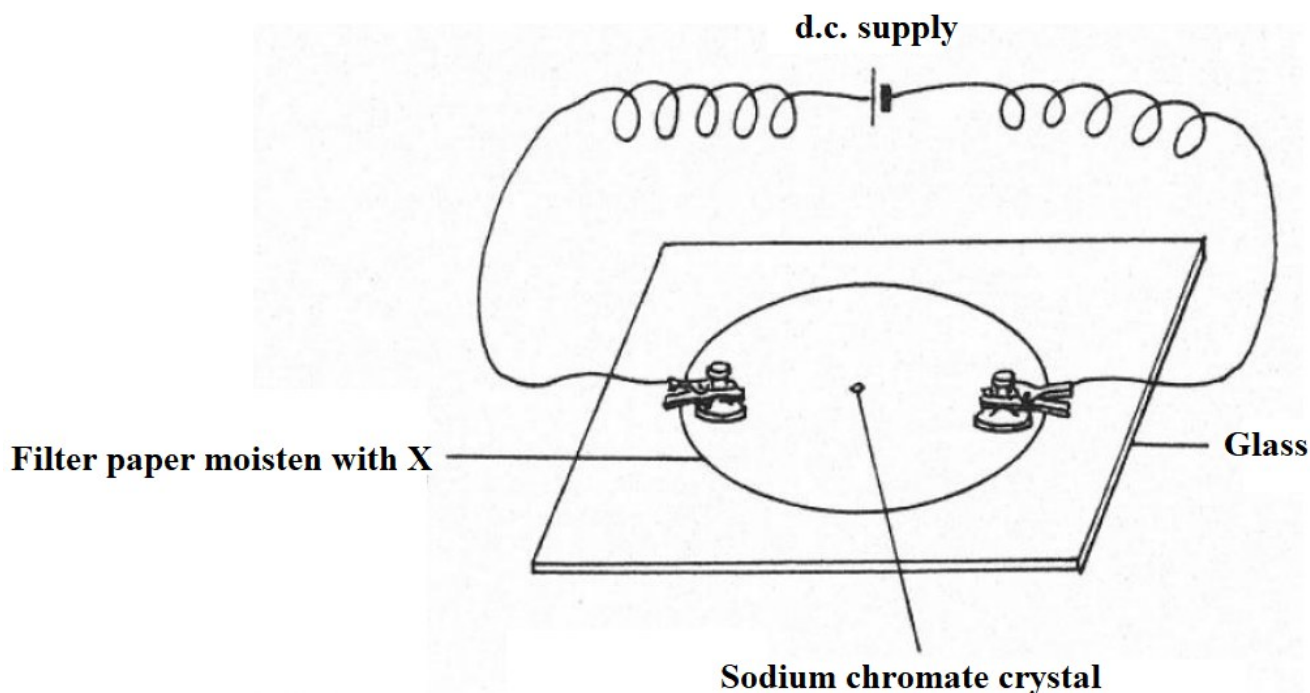
(1 mark)

37. M is a metal. It forms an ionic compound with chlorine which has the formula of MCl_2 .
- Write the electronic arrangements of a chlorine atom and a chloride ion respectively. (2 marks)
 - What is the charge on an ion of M in MCl_2 ? (1 mark)
 - An ion of M in MCl_2 has 34 neutrons and 27 electrons. With reference to the Periodic Table, suggest what M is. (1 mark)
 - Predict the colour of an aqueous solution of MCl_2 . (1 mark)
38. The set-up shown in the diagram below is used to study the migration of ions. After the circuit is closed for some time, a purple spot is found in the position of the filter paper as shown below.



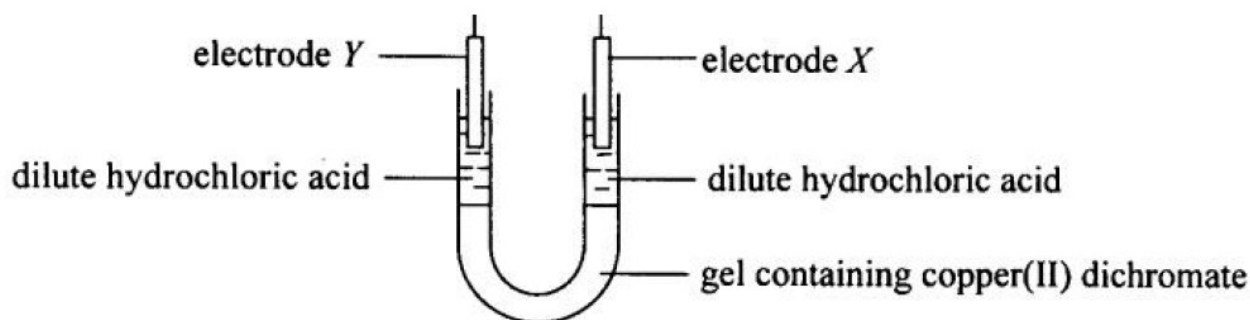
- Why is the filter paper soaked with sodium sulphate solution instead of water? (1 mark)
- Which ion is responsible for the purple spot? (1 mark)
- Which electrode, X or Y, is the positive electrode? Explain your answer. (1 mark)
- If the positive and negative electrodes in the above set-up are reversed, what would be observed? Explain your answer (2 marks)

39. Consider the experimental set-up below:



- (a) Write down the chemical formula of sodium chromate. (1 mark)
- (b) Write down the electronic arrangement of sodium ion. (1 mark)
- (c) What is X? What was the purpose of moistening the filter paper with X? (2 marks)
- (d) After few minutes, a coloured patch was seen to develop on the filter paper.
 - (i) What is the colour of the patch? (1 mark)
 - (ii) What ion was responsible for the coloured patch? (1 mark)
 - (iii) Sketch the position where the coloured patch would be expected to develop. (1 mark)
- (e) If the above experiment had been in progress for some time and the poles of d.c. supply were then reversed, briefly describe what would happen. (1 mark)
- (f) Suggest one way to speed up the experiment. (1 mark)

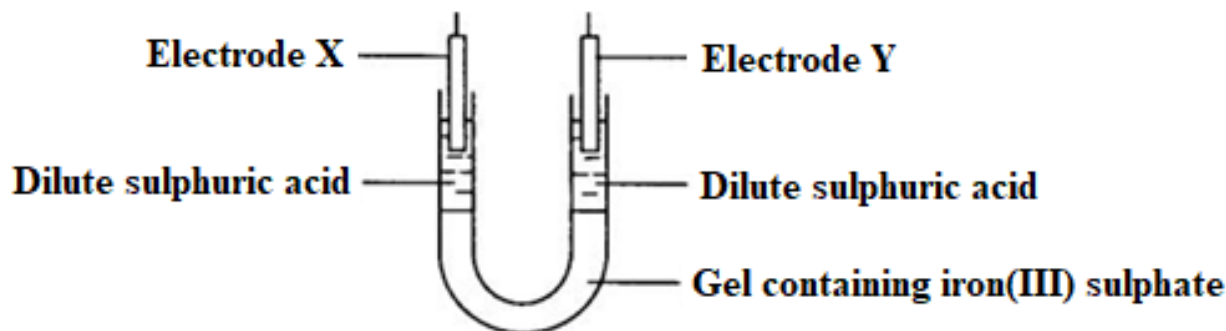
40. The diagram below shows the experimental set-up of investigating the migration of colored ion.



After the electric current has passed for 10 minutes, it was found that blue color gradually appears in the solution around electrode Y.

- (a) Write the chemical formula of copper(II) dichromate (1 mark)
- (b) The diagram above is incomplete. Complete the diagram above and provide the missing label. (1 mark)
- (c) What will be observed in electrode X? Explain your answer. (2 marks)
- (d) Explain why the migration of colored ions CANNOT be observed if copper(II) dichromate is replaced by calcium chloride (1 mark)
- (e) What will be the difference in observation if copper(II) dichromate is replaced by potassium permanganate in the above set-up? (2 marks)
- (f) Why is a gel containing copper(II) dichromate, rather than an aqueous solution of the compound, used in this experiment? (2 marks)

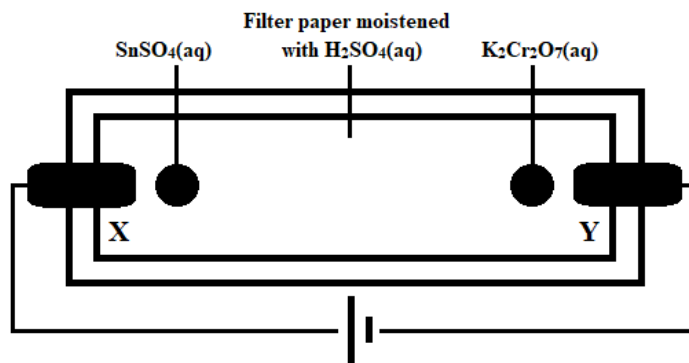
41. The diagram below shows the experimental set-up of investigating the migration of colored ion.



After the electric current has passed for 10 minutes, it was found that yellow color gradually appears in the solution around electrode Y.

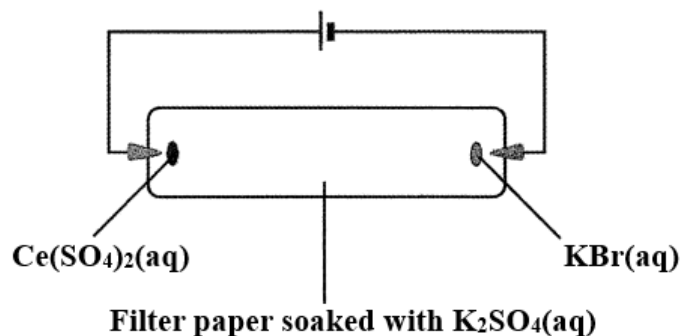
- (a) Write the chemical formula of iron(III) sulphate (1 mark)
- (b) The diagram above is incomplete. Complete the diagram above and provide the missing label. (1 mark)
- (c) State the color change of solution, if any, around electrode X. Explain your answer. (2 marks)
- (e) Explain why the migration of colored ion CANNOT be observed if iron(III) sulphate is replaced by ammonium sulphate. (1 mark)
- (f) Why is a gel containing copper(II) dichromate, rather than an aqueous solution of the compound, used in this experiment? (2 marks)

42. Consider the following experimental set-up:



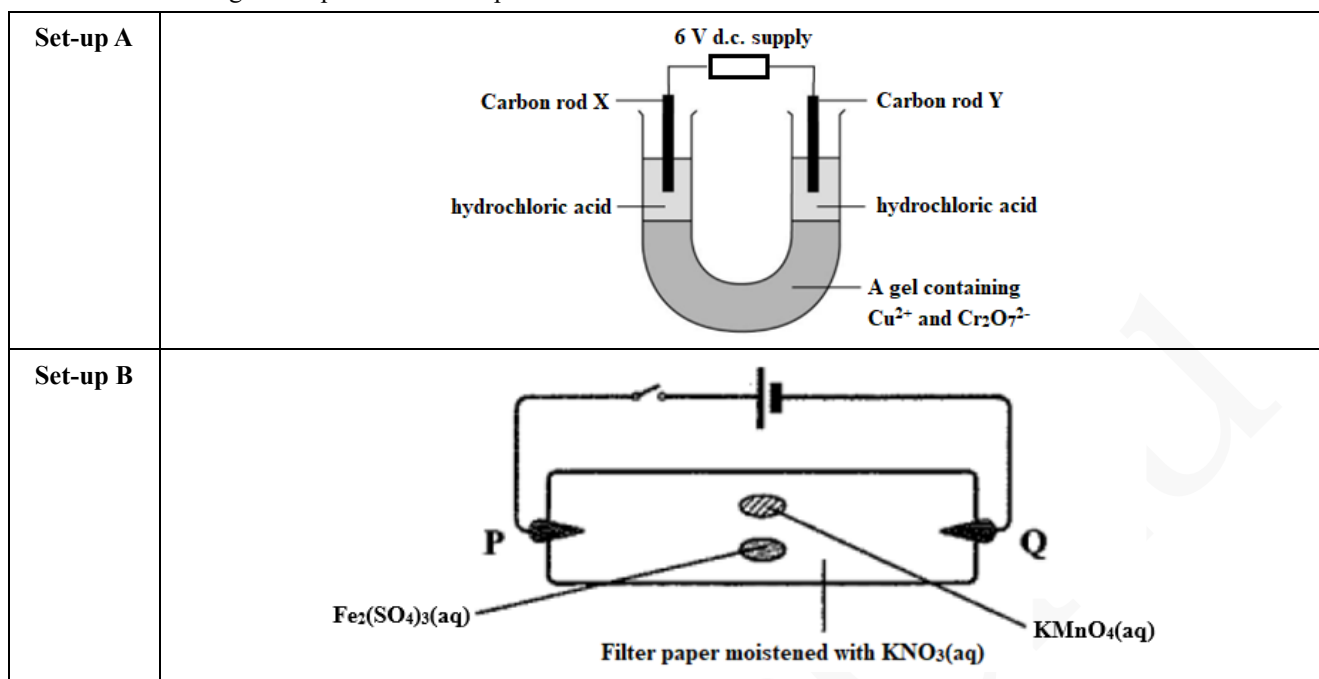
- (a) The filter paper used should be moistened with dilute sulphuric acid ($\text{H}_2\text{SO}_4(\text{aq})$).
- (i) What is the function of dilute sulphuric acid? (1 mark)
- (ii) Explain why the filter paper cannot be moistened with ethanol. (1 mark)
- (b) What is the chemical name of $\text{K}_2\text{Cr}_2\text{O}_7$? (1 mark)
- (c) Tin(II) sulphate (SnSO_4) is an ionic compound which is soluble in water. Tin(II) ion (Sn^{2+}) can react with $\text{Cr}_2\text{O}_7^{2-}$ in the presence of acid. The equation below shows the reaction between $\text{Sn}^{2+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ in the presence of acid:
- $$3\text{Sn}^{2+}(\text{aq}) + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) \rightarrow 3\text{Sn}^{4+}(\text{aq}) + 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$$
- (i) It is given that both $\text{Sn}^{2+}(\text{aq})$ and $\text{Sn}^{4+}(\text{aq})$ are colorless. Explain what would be observed around the middle of the filter paper when the circuit is closed for a period of time. (2 marks)
- (ii) The experiment is repeated but the two poles of the cell have been reversed at the very beginning. Explain what would be observed around the middle of the filter paper when the circuit is closed for a period of time. (4 marks)
- (d) Suggest one method to speed up the experiment (1 mark)

43. The set-up of an experiment for studying the movement of ions is shown below.



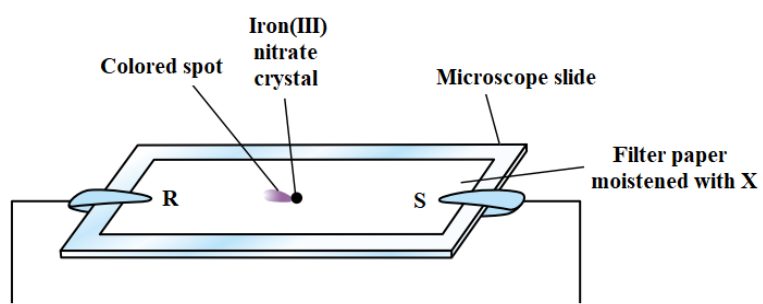
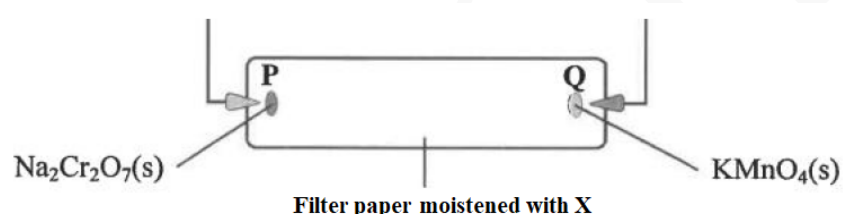
- (a) (i) State the purpose of soaking the filter paper with $\text{K}_2\text{SO}_4(\text{aq})$. (1 mark)
- (ii) Explain whether the filter paper can be soaked with $\text{KMnO}_4(\text{aq})$ instead of $\text{K}_2\text{SO}_4(\text{aq})$. (1 mark)
- (b) Cerium(IV) ion (Ce^{4+}) can react with bromide ion (Br^-) in the equation below:
- $$2\text{Ce}^{4+}(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Ce}^{3+}(\text{aq}) + \text{Br}_2(\text{aq})$$
- (i) It is given that $\text{Ce}^{3+}(\text{aq})$ is colorless. Explain what would be observed around the middle of the filter paper when the circuit is closed for a period of time. (2 marks)
- (ii) The experiment is repeated, but the $\text{KBr}(\text{aq})$ is replaced by $\text{KI}(\text{aq})$ at the very beginning. It is expected that the observation will be similar to observation in (b)(i). Suggest a reason for this expectation based on electronic arrangement. (1 mark)
- (c) The experiment is repeated, but the two poles of the cell have been reversed at the very beginning. Explain what would be observed around the middle of the filter paper when the circuit is closed for a period of time. (2 marks)

44. Consider the following two experimental set-ups:



- (a) In set-up A, a blue colour can be observed around graphite electrode X when an electric current is passed through the circuit for some time.
- Explain why dilute hydrochloric acid is added to each side of the tube. (1 mark)
 - Deduce the polarity of carbon rod X. Explain your answer. (2 marks)
 - What would be the colour observed around carbon rod Y? Explain your answer. (2 marks)
 - What would happen if the carbon rods were replaced by wooden rods? Explain your answer. (1 mark)
- (b) Consider set-up B:
- Explain whether the filter paper can be soaked with $\text{Ni}(\text{NO}_3)_2(\text{aq})$ instead of $\text{KNO}_3(\text{aq})$. (1 mark)
 - State what would be observed on the filter paper when the circuit is closed for a period of time. (2 marks)
 - The experiment is repeated, but the two poles of the cell have been reversed at the very beginning. Suggest what would be observed on the filter paper when the circuit is closed for a period of time. (1 mark)

45. Migration of colored ions can be observed in the following two experimental set-ups.

Set-up A	
Set-up B	

- (a) What is X? Suggest why it is added in both experimental set-ups. (2 marks)
- (b) Suggest one method to speed up the experiment. (1 mark)
- (c) Consider experimental **set-up A**:
- Give the chemical formula of iron(III) nitrate (1 mark)
 - Complete the diagram above. (1 mark)
 - What will be the observation if the experimental set-up is repeated WITHOUT connecting to the power supply? (1 mark)
- (d) Consider experimental **set-up B**:
- Give the chemical names of $\text{Na}_2\text{Cr}_2\text{O}_7$ and KMnO_4 (2 marks)
 - After the electric current has passed for 10 minutes, purple color patch can be found in the middle of the filter paper. Complete the diagram above. (1 mark)
 - What will be observed if electrode P and electrode Q are interchanged, while other experimental conditions remain unchanged? Explain your answer. (2 marks)

PERIODIC TABLE 週期表

GROUP 族

atomic number 原子序

relative atomic mass 相對原子質量

		I		II												0					
3	Li	4	Be			21	22	23	24	25	26	27	28	29	30	5	6	7	8	9	10
6.9						Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	B	C	N	O	F	Ne
11	Na	12	Mg													13	14	15	16	17	18
23.0																Al	Si	P	S	Cl	Ar
24.3																27.0	28.1	31.0	32.1	35.5	40.0
19	K	20	Ca			39	40	41	42	43	44	45	46	47	48	31	32	33	34	35	36
						Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Ga	Ge	As	Se	Br	Kr
39.1										(98)	101.1	102.9	106.4	107.9	112.4	69.7	72.6	74.9	79.0	79.9	83.8
37	Rb	38	Sr			57 *										49	50	51	52	53	54
85.5						La										In	Sn	Sb	Te	I	Xe
55	Cs	56	Ba			138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	114.8	118.7	121.8	127.6	126.9	131.3
132.9						89 **										81	82	83	84	85	86
																Tl	Pb	Bi	Po	At	Rn
87	Fr	88	Ra			104	105									204.4	207.2	209.0	(209)	(210)	(222)
(223)																					
(226)																					

1

H

1.0

relative atomic mass 相對原子質量

		58	59	60	61	62	63	64	65	66	67	68	69	70	71
		Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
		140.1	140.9	144.2	(145)	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
		90	91	92	93	94	95	96	97	98	99	100	101	102	103
		Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
		232.0	(231)	238.0	(237)	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

*

**

Part 4

1. Draw the electron diagram for each of the following molecules (showing electrons in the outermost shells only).

(a) H_2	(b) O_2
(c) Cl_2	(d) N_2
(e) H_2O	(f) HF
(g) NH_3	(h) CH_4
(i) CS_2	(j) OF_2

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2. Draw the electron diagram for each of the following molecules (showing electrons in the outermost shells only).

(a) Hydrogen and Oxygen	(b) Oxygen and Fluorine
(c) Carbon and Chlorine	(d) Hydrogen and Nitrogen
(e) Sulphur and Carbon	(f) Sulphur and Fluorine
(g) Silicon and Chlorine	(h) Phosphorus and Fluorine

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3. Draw the electron diagram for each of the following molecules (showing electrons in the outermost shells only).

(a) HCN	(b) NOF
(c) CH ₂ O	(d) N ₂ F ₂
(e) N ₂ H ₄	(f) O ₂ F ₂
(g) C ₂ H ₂	(h) S ₂ Cl ₂

4. (a) Draw the electron diagram of the following molecules, showing electrons in the outermost shells only.

(i) Nitrosyl chloride (NOCl)	(ii) Cyanogen chloride (CNCl)
(iii) Dioxygen difluoride (O ₂ F ₂)	(iv) Dichloroacetylene (C ₂ Cl ₂)
(v) Boron trichloride (BCl ₃)	(vi) Phosgene (COCl ₂)
(Hint: BCl ₃ does NOT have an octet structure)	

(6 marks)

- (b) Which of the above molecules has the greatest number of **lone pairs of electrons**? (1 mark)
- (c) Which of the above molecules has the greatest number of **electrons**? (1 mark)
- (d) Which of the above molecules has / have **double covalent bond** in their molecules? (1 mark)
- (e) Which of the above molecules has / have **triple covalent bond** in their molecules? (1 mark)

5. (a) Draw the electron diagram of the following compounds, showing electrons in outermost shells only.

(i) CO_2	(ii) H_2O
(iii) NH_3	(iv) CH_4
(v) H_2O_2	(vi) N_2H_2
(vii) C_2Cl_2	(viii) NOCl
(ix) COCl_2	(x) C_2H_4

(10 marks)

- (b) Which of the compounds in (a):

(i) contain(s) double bond?

(1 mark)

(ii) contain(s) triple bond?

(1 mark)

(iii) do(es) not contain lone pair?

(1 mark)

(iv) have 6 lone pairs of electrons (i.e. 12 lone pair electrons)

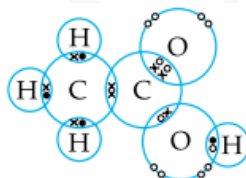
(1 mark)

6. Consider the elements shown below:

Element	X	Y	Z
Atomic number	8	9	14

- (a) Draw the electron diagram of a molecule of X, showing electrons in the outermost shells only. (1 mark)
- (b) What is the type of chemical bond that joins the atoms together in a molecule of X? (1 mark)
- (c) Compound A forms when Y reacts with Z.
- (i) Write the electronic arrangement of an atom of Y and that of an atom of Z. (2 marks)
- (ii) Draw the electron diagram of a molecule of A, showing electrons in the outermost shells only. (1 mark)
- (iii) State the number of bond pairs and that of lone pairs in a molecule of A. (2 marks)

7. The electron diagram of a molecule of ethanoic acid, which is the main ingredient of vinegar, is shown below.



- (a) Describe how a C–H bond in ethanoic acid forms. (1 mark)
- (b) How many atoms do the left carbon atom and the right carbon atom bond to? (2 marks)
- (c) Between which two atoms is there a double covalent bond? (1 mark)
- (d) Write the structural formula and molecular formula of ethanoic acid. (2 marks)

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8. Consider the four elements W, X, Y and Z. (Note: the letters are NOT the symbols of the elements.)

Element	W	X	Y	Z
Atomic number	12	15	17	18

- (a) Two of these elements can combine to form a covalent compound. Give the formula of the compound formed. (1 mark)
- (b) Describe the bond formation in the covalent compound in (a) from atoms of the relevant elements. (2 marks)
- (c) Draw the electron diagram for a molecule of the compound in (a), showing electrons in the outermost shells only. (1 mark)
9. The non-metallic element selenium (Se) has six outermost shell electrons in its atom.
- (a) To which group in the periodic table does selenium belong? (1 mark)
- (b) Hydrogen selenide is a compound made up of hydrogen and selenium.
- (i) Predict the type of bonding present in hydrogen selenide. (1 mark)
- (ii) Draw an electron diagram of hydrogen selenide, showing electrons in the outermost shells only. (1 mark)

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10. Both nitrogen and neon are gases under room conditions.

(a) Write the electronic arrangement of a neon atom.

(1 mark)

(b) Draw the electron diagram of a nitrogen molecule, showing electrons in the outermost shells only.

(1 mark)

(c) Nitrogen exists as diatomic molecules while neon exists as monoatomic molecules.

(i) Explain why monoatomic neon is stable.

(1 mark)

(ii) Explain why diatomic nitrogen is more stable than monoatomic nitrogen.

(2 marks)

11. The table below gives the melting points of Group VII elements.

Element	Atomic number	Melting point / °C
Fluorine	9	-220
Chlorine	17	-101
Bromine	35	-7
Iodine	53	113
Astatine	85	302

(a) What is the name given to Group VII elements of the periodic table?

(1 mark)

(b) (i) Explain why elements in Group VII exist as diatomic molecules.

(1 mark)

(ii) Name TWO other elements, apart from those in Group VII, which are diatomic.

(2 marks)

(c) Draw an electron diagram of the compound formed from fluorine and chlorine, showing electrons in the outermost shells only.

(1 mark)

(d) Draw an electron diagram of the compound formed from chlorine and oxygen, showing electrons in the outermost shells only.

(1 mark)

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12. Give chemical formulae to the following covalent compounds

	Chemical name	Chemical formulae
(a)	Carbon dioxide	
(b)	Phosphorus trichloride	
(c)	Chlorine dioxide	
(d)	Carbon monoxide	
(e)	Phosphorus pentachloride	
(f)	Sulphur trioxide	
(g)	Oxygen difluoride	
(h)	Chlorine trifluoride	
(i)	Dinitrogen tetroxide	
(j)	Dinitrogen pentoxide	
(k)	Water	
(l)	Ammonia	
(m)	Methane	
(n)	Hydrogen peroxide	
(o)	Hydrogen chloride	
(p)	Hydrogen sulphide	

13. Give chemical formulae to the following covalent compounds

	Chemical formulae	Chemical name
(a)	NO	
(b)	SF ₆	
(c)	CO	
(d)	CO ₂	
(e)	PCl ₅	
(f)	NCl ₃	
(g)	P ₂ O ₅	
(h)	S ₂ Cl ₂	
(i)	O ₂ F ₂	
(j)	N ₂ F ₄	
(k)	N ₂ O	
(l)	NH ₃	
(m)	CH ₄	
(n)	H ₂ O ₂	
(o)	H ₂ O	
(p)	HF	

15. Complete the following table

Chemical name	Chemical formula
Potassium sulphate	
	CO ₂
Sodium hydroxide	
	Ca(NO ₃) ₂
Sulphur hexafluoride	
	ZnSO ₄
Iron(III) chloride	
	CuSO ₄
Phosphorus trichloride	
	K ₂ SO ₃
Aluminum hydroxide	
	N ₂ O ₅
Sulphur trioxide	
	K ₂ Cr ₂ O ₇
Ammonium carbonate	
	N ₂ O ₅
Lithium nitride	
	K ₂ S
Iron(III) oxide	
	NaNO ₂
Phosphorus tribromide	
	PbO
Calcium carbonate	
	Mg ₃ N ₂
Iron(III) sulphate	
	H ₂ O
Nitrogen dioxide	
	NH ₄ NO ₃
Magnesium phosphate	
	SiF ₄
Ammonia	
	S ₂ Cl ₂
Calcium hydrogencarbonate	
	FePO ₄
Lead(II) carbonate	

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16. Complete the following table:

Chemical name	Chemical formula	Formula mass
Carbon dioxide		
Water		
Sodium chloride		
Sodium carbonate		
Potassium nitrate		
Nitrogen dioxide		
Ammonia		
Ammonium sulphate		
Sulphur trioxide		
Magnesium hydroxide		
Sodium phosphate		
Aluminium chloride		
Copper(II) sulphate		
Iron(III) oxide		
Iron(II) hydroxide		
Carbon disulphide		
Dinitrogen pentoxide		
Lithium carbonate		
Calcium hydrogencarbonate		
Potassium sulphite		
Potassium permanganate		
Phosphorus pentachloride		
Ammonium dichromate		
Iron(III) sulphate		
Zinc nitrate		
Sodium nitrite		
Potassium phosphate		
Sulphur hexafluoride		
Copper(I) oxide		
Magnesium nitride		
Hydrogen peroxide		

17. Complete the following table. (Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, F = 19.0, S = 32.1, Cl = 35.5)

(a)	Electron diagram	Chemical name
		Nitrogen
		Molecular formula
		Relative molecular mass
(b)	Electron diagram	Chemical name
		Chlorine monofluoride
		Molecular formula
		Relative molecular mass
(c)	Electron diagram	Chemical name
		Hydrogen sulphide
		Molecular formula
		Relative molecular mass
(d)	Electron diagram	Chemical name
		Molecular formula
		H ₂ O ₂
		Relative molecular mass
(e)	Electron diagram	Chemical name
		Molecular formula
		N ₂ F ₂
		Relative molecular mass

(15 marks)

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18. Draw the **structural formula** for each of the following molecules (showing electrons in the outermost shells only).

(a) Oxygen	(b) Nitrogen
(c) Oxygen dichloride	(d) Nitrogen trichloride
(e) Methane	(f) Ammonia
(g) Water	(h) Hydrogen peroxide

19. Ammonium nitrate will be decomposed upon heating to give oxygen, nitrogen and water

(a) Complete the following table

Molecule	Electron diagram	Structural formula	Atomicity	Number of lone pairs in one molecule
Nitrogen				
Oxygen				
Water				

(3 marks)

(b) State ALL attractive forces that can be found in solid ammonium nitrate.

(1 mark)

(c) Calculate the formula mass of the following substances. [Relative atomic masses: H = 1.0, N = 14.0, O = 16.0]

(i) Ammonium nitrate

(1 mark)

(ii) Water

(1 mark)

(iii) Oxygen gas

(1 mark)

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20. Draw the **electron diagrams** (showing electrons in the outermost shell only) of the compounds formed between the following pairs of elements. Give the name of the compound formed.

(a) Sodium + Oxygen	(b) Magnesium + Chlorine
Name of compound:	Name of compound:
(c) Carbon + Sulphur	(d) Fluorine + Calcium
Name of compound:	Name of compound:
(e) Chlorine + Phosphorus	(f) Aluminium + Oxygen
Name of compound:	Name of compound:
(g) Sulphur + Calcium	(h) Potassium + Nitrogen
Name of compound:	Name of compound:
(i) Carbon + Hydrogen	(j) Chlorine + Oxygen
Name of compound:	Name of compound:
(k) Nitrogen + Magnesium	(l) Hydrogen + Nitrogen
Name of compound:	Name of compound:

21. Magnesium and chlorine can react with oxygen to give magnesium oxide and dichlorine monoxide respectively.
- (a) Consider magnesium oxide:
- (i) Draw the electron diagram of magnesium oxide, showing electrons in the outermost shells only. (1 mark)
- (ii) What is the color of solid magnesium oxide? (1 mark)
- (iii) Magnesium also reacts with oxygen under suitable conditions to give magnesium peroxide. Given that the formula of peroxide ion is O_2^{2-} . Calculate the formula mass of magnesium peroxide.
(Relative atomic masses: $\text{O} = 16.0$, $\text{Mg} = 24.3$) (2 marks)
- (b) Consider dichlorine monoxide.
- (i) Draw the electron diagram of dichlorine monoxide, showing electrons in the outermost shells only. (1 mark)
- (ii) Chlorine also reacts with oxygen under suitable conditions to give dichlorine heptoxide. Write down the molecular formula of dichlorine heptoxide. (1 mark)
- (c) Magnesium also reacts with nitrogen to form magnesium azide. The formula of azide ion is N_3^- .
- (i) Write down the chemical formula of magnesium azide. (1 mark)
- (ii) Calculate the formula mass of magnesium azide.
(Relative atomic masses: $\text{N} = 14.0$, $\text{Mg} = 24.3$) (1 mark)

22. Carbon and nitrogen are Period 2 elements in the Periodic Table. They can form various compounds.
- (a) Cyanogen (C_2N_2) is a molecule formed between carbon and nitrogen.
- (i) Draw the electron diagram of a cyanogen molecule, showing electrons in the outermost shells only. (1 mark)
- (ii) How many lone pair(s) is / are present in a cyanogen molecule? (1 mark)
- (b) Glyoxal is a carbon containing compound with molecular formula $\text{C}_2\text{H}_2\text{O}_2$.
- (i) It is given that a glyoxal molecule contains one carbon-carbon single bond. Draw the electron diagram of a glyoxal molecule, showing electrons in the outermost shells only. (1 mark)
- (ii) How many bond pair(s) is / are present in a glyoxal molecule? (1 mark)
- (c) When lithium metal is burnt in air, nitrogen in air may also react with lithium to form lithium nitride which is an ionic compound.
- (i) Draw the electron diagram of lithium nitride, showing ELECTRONS IN ALL SHELLS. (1 mark)
- (ii) Lithium carbide is an ionic compound formed between lithium and carbon.
- (1) It is given that the formula of a carbide ion is C_2^{2-} . Calculate the formula mass of lithium carbide. (Relative atomic masses: $\text{Li} = 6.9$, $\text{C} = 12.0$) (1 mark)
- (2) When lithium carbide is added to water, acetylene gas (C_2H_2) is produced. Draw the electron diagram of an acetylene molecule, showing electrons in the outermost shells only. (1 mark)
- (d) Nitrogen can form different compounds with hydrogen. Draw the structural formula for each of the following compounds formed between nitrogen and hydrogen.
- (i) Ammonia (1 mark)
- (ii) Diazene (N_2H_2) (1 mark)
- (iii) Hydrazine (N_2H_4) (1 mark)

23. Consider the table below:

Element	Atomic number	Electronic arrangement
A	1	1
B	2	2
C	6	
D	7	
E	8	
F	11	
G	19	
H	20	
I		2,8,18,7
J		2,8,18,18,8

- (a) Complete the above table (8 marks)
- (b) Which of the above elements seldom form(s) compounds with other elements? (1 mark)
- (c) Which of the above elements is/are non-metal? (1 mark)
- (d) Draw the electron diagram (showing outermost shell only) of the compounds formed between:
- (i) A and G (1 mark)
- (ii) A and I (1 mark)
- (iii) D and H (1 mark)
- (iv) D and I (1 mark)
- (v) C and E (1 mark)
- (vi) E and H (1 mark)

24. Consider the table below:

Element	Description	Name of element
A	Element A is a non-metal which can conduct electricity	
B	Atom of element B has 2 fully filled electron shells	
C	Element C is a metal which is a liquid at room temperature	
D	Element D is a non-metal which is a liquid at room temperature	
E	Element E is a metalloid. Sulphur and element E are elements that placed in the same period in the Periodic Table.	
F	Element F does not belong to any group in the periodic table	
G	Atomic number of element G is 2	
H	An atom of element H can form a stable ion by losing 2 electrons. The stable ion formed by element H and an atom of element B have the same electronic arrangement	
I	An atom of element I can form a stable ion by gaining 3 electrons. The stable ion formed by element I and an atom of element B have the same electronic arrangement	
J	In an atom of J with isotopic mass of 35, there are 18 neutrons in it.	

(a) Complete the above table.

(10 marks)

(b) Draw the electron diagram (showing outermost shell only) of the compounds formed between:

(i) A and D

(1 mark)

(ii) D and E

(1 mark)

(iii) D and H

(1 mark)

(iv) H and I

(1 mark)

(v) F and I

(1 mark)

(vi) D and F

(1 mark)

25. The table below shows some information of three atoms

	Number of protons	Number of electrons	Number of neutrons
Protium	1	1	0
Deuterium	1	1	1
Oxygen			8

- (a) Complete the above table (1 mark)
- (b) Give the full atomic symbol of a deuterium atom. (1 mark)
- (c) Name the relationship between protium and deuterium. Explain your answer. (2 marks)
- (d) Deuterium can be represented by D. It reacts with oxygen as shown in the equation below:
- $$2\text{D}_2 + \text{O}_2 \rightarrow 2\text{D}_2\text{O}$$
- (i) Draw the electron diagram for a D_2O molecule, showing electrons in the outermost shell only (1 mark)
- (ii) A small piece of potassium metal is placed into $\text{D}_2\text{O}(\text{l})$ at room conditions. State TWO expected observations (2 marks)
- (e) Calculate the formula mass of a D_2O molecule. (1 mark)

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26. The following is a part of the Periodic Table which shows only six of the elements A to F. (Note: A to F are NOT the symbols of the elements.)

Group \ Period	I	II	III	IV	V	VI	VII	0
2	A			B		C		D
3		E					F	

- (a) In which element(s) do their atoms / does its atoms have an octet electronic arrangement? (1 mark)
- (b) Draw electron diagrams (showing electrons in the outermost shells only) for the compounds formed between
- (i) elements A and F. (1 mark)
- (ii) elements C and E. (1 mark)
- (iii) elements B and C. (1 mark)
- (c) Draw electron diagrams (showing electrons in the outermost shells only) for
- (i) a molecule of C. (1 mark)
- (ii) a molecule of F. (1 mark)

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27. The following table lists the atomic numbers of four elements.

Element	W	X	Y	Z
Atomic number	7	9	10	20

- (a) Suggest why element Y does NOT form compounds. (1 mark)
- (b) P is a compound formed from elements W and X.
- (i) Predict the type of compound (ionic or covalent) formed. (1 mark)
- (ii) Draw an electron diagram of P, showing electrons in the outermost shells only. (1 mark)
- (iii) State the chemical formula of P. (1 mark)
- (c) Q is a compound formed from elements X and Z.
- (i) Predict the type of compound (ionic or covalent) formed. (1 mark)
- (ii) Draw an electron diagram of Q, showing electrons in the outermost shells only. (1 mark)
- (iii) State the chemical formula of Q. (1 mark)

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28. X, Y and Z are the first 20 elements in the periodic table, with the sum of their atomic numbers equal to 46. Moreover, both X and Y are Group VII elements, while the atomic number of X is smaller than that of Y.
- (a) What are elements X, Y and Z? (3 marks)
- (b) Give the special name of the group to which Z belongs. (1 mark)
- (c) Draw an electron diagram of the compound P formed from X and Y, showing electrons in the outermost shells only. (1 mark)
- (d) Draw an electron diagram of the compound Q formed from X and Z, showing electrons in the outermost shells only. (1 mark)
- (e) X combines with nitrogen to form a nitride. Draw an electron diagram of the nitride, showing electrons in the outermost shells only. (1 mark)
- (f) Z combines with nitrogen to form a nitride. Draw an electron diagram of the nitride, showing electrons in the outermost shells only. (1 mark)

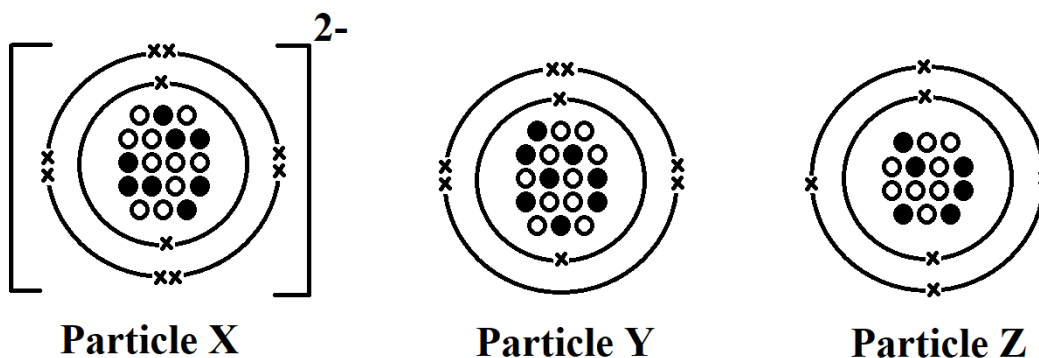
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29. The following table gives some information about V, W, X, Y and Z which represent atoms of some elements. It is given that V and W are isotopes.

	Mass number	Atomic number	No. of proton	No. of neutron	No. of electron
V	35				
W			17	20	
X	40	20			
Y				16	15
Z	22			12	

- (a) Complete the above table. (5 marks)
- (b) What is the name of the group that Z belongs to in the Periodic Table? (1 mark)
- (c) It is given that V and W are chlorine atoms. The relative atomic mass of chlorine is 35.5.
- (i) What is meant by the term 'relative atomic mass'? (1 mark)
- (ii) It is given that chlorine occurs naturally in two isotopes, V and W. Calculate the abundance of V. (2 marks)
- (iii) Explain whether V and W can be separated by chemical means. (1 mark)
- (d) Compound M is formed when V reacts with X.
- (i) Draw the electron diagram of M, showing electrons in the outermost shells only. (1 mark)
- (ii) Calculate the formula mass of M. (1 mark)
- (e) Compound N is formed when W reacts with Y.
- (i) Draw the electron diagram of N, showing electrons in the outermost shells only. (1 mark)
- (ii) Calculate the formula mass of N. (1 mark)

30. The diagram below shows the structures of particle X, particle Y and particle Z.

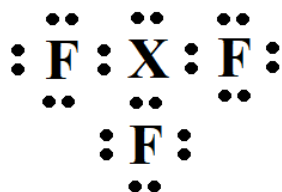


- (a) Name the following particles.
- Black dot (1 mark)
 - White dot (1 mark)
 - Cross (1 mark)
- (b) Draw the electron diagram for a MOLECULE of element of particle Y, showing electrons in the outermost shells only. (1 mark)
- (c) Draw the electron diagram for the compound formed between element of particle X and element of particle Z, showing electrons in outermost shells only. (1 mark)
- (d) What is the relationship between particle X and particle Y? (1 mark)
- (e) The table below shows some information of element P and Q.

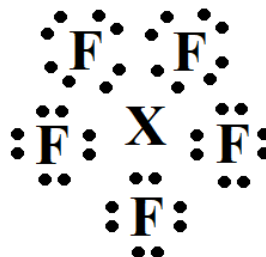
	P	Q
Atomic number	16	34
Number of occupied electron shells in the atoms		
Number of electrons in the outermost shell in the atoms		

- Complete the above table with reference to the Periodic Table (2 marks)
- State the electronic arrangement of an atom of Q (1 mark)
- Element P can be used to treat spilled mercury. It is expected that element Q can also be used. Suggest a reason for this expectation based on electronic arrangement. (1 mark)

31. Element X can combine with fluorine to give TWO different compounds. The electron diagrams of two compounds formed (compound A and compound B) are shown below:



Compound A



Compound B

It is known that element X is a period 3 element.

- (a) Name element X. Hence name compound B. (2 marks)
- (b) Element X can react with sodium to form compound K.
 - (i) Give the formula of compound K. (1 mark)
 - (ii) Draw the electron diagram of compound K, showing electrons in the outermost shells only. (1 mark)
 - (iii) What attractions can be found between the particles of compound K? (1 mark)
 - (iv) Tom claims that all particles in compound K have the electronic arrangement of argon. Do you agree? Explain your answer. (1 mark)
- (c) State the number of protons in a molecules of compound A. (2 marks)

32. Dative covalent bond is present in some chemicals.

(a) H_2O molecules can react with H^+ to form H_3O^+ ions.

(i) In each of the following boxes, draw the electron diagram (showing ELECTRONS IN THE OUTERMOST SHELLS only) for a suitable chemical species to show the formation of a H_3O^+ ion.



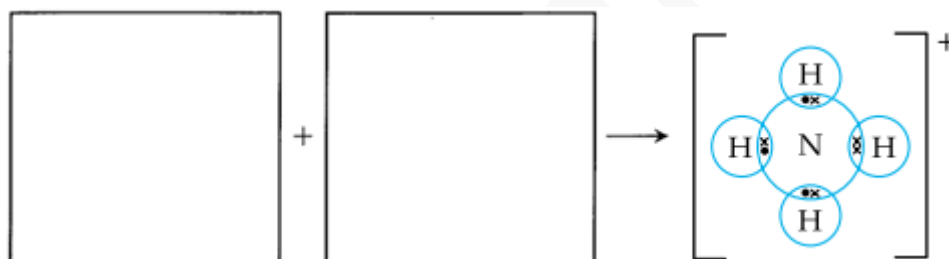
(2 marks)

(ii) Describe the formation of dative covalent bond using H_3O^+ as an example.

(1 mark)

(b) NH_3 molecules can react with H^+ to form NH_4^+ ions.

(i) In each of the following boxes, draw the electron diagram (showing ELECTRONS IN THE OUTERMOST SHELLS only) for a suitable chemical species to show the formation of a NH_4^+ ion.



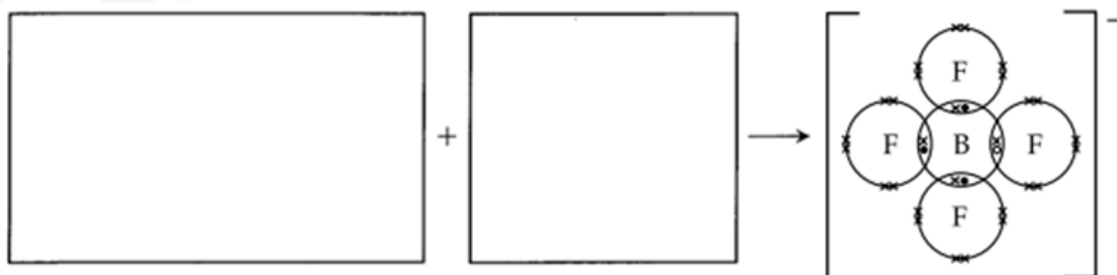
(2 marks)

(ii) Describe the formation of dative covalent bond using NH_4^+ as an example.

(1 mark)

(c) BF_3 molecules can react with F^- to form BF_4^- ions

(i) In each of the following boxes, draw the electron diagram (showing ELECTRONS IN THE OUTERMOST SHELLS only) for a suitable chemical species to show the formation of a BF_4^- ion.



(2 marks)

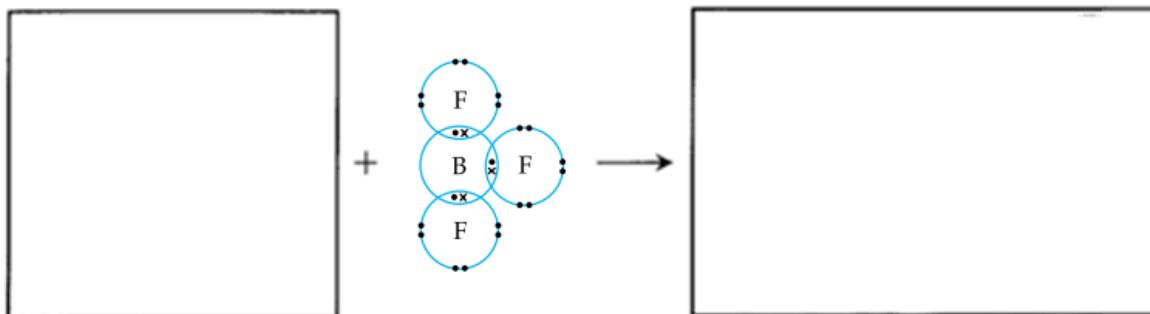
(ii) Describe the formation of dative covalent bond using BF_4^- as an example.

(1 mark)

33. Dative covalent bond is present in some chemicals.

(a) BF_3 molecules can form H_3NBF_3 .

(i) In each of the following boxes, draw the electron diagram (showing ELECTRONS IN THE OUTERMOST SHELLS only) for a suitable chemical species to show the formation of a H_3NBF_3 .



(ii) Write the structural formula for a H_3NBF_3 molecule, using an arrow ($\rightarrow$) to represent the dative covalent bond.

(1 mark)

(b) Oxygen molecule can react with an oxygen atom to give an ozone molecule (O_3).

(i) Describe the bond formation between oxygen molecule and oxygen atom.

(3 marks)

(ii) Write the structural formula for an ozone molecule, using an arrow ($\rightarrow$) to represent the dative covalent bond.

(1 mark)

(c) Nitrogen molecule can react with an oxygen atom to give a nitrous oxide molecule (N_2O).

(i) Describe the bond formation between nitrogen molecule and oxygen atom.

(3 marks)

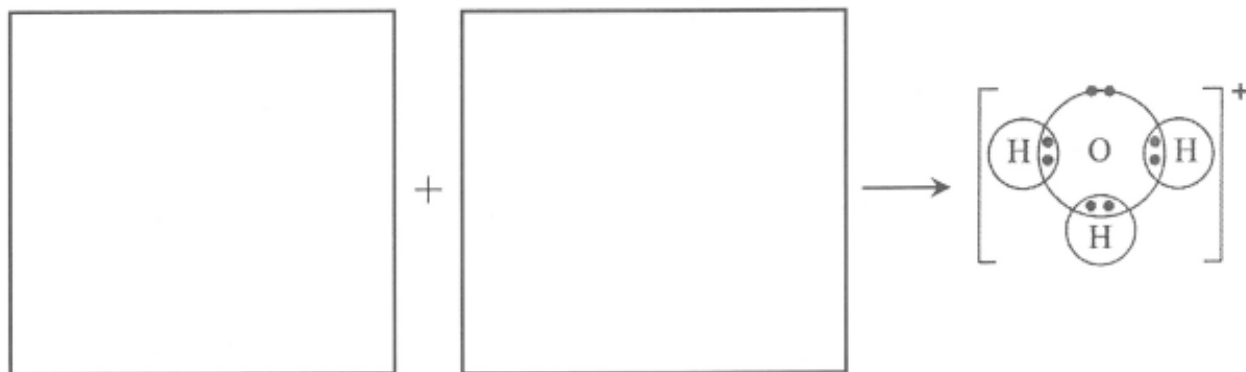
(ii) Write the structural formula for a nitrous oxide molecule, using an arrow ($\rightarrow$) to represent the dative covalent bond.

(1 mark)

34. (a) Describe and explain the formation of dative covalent bond in the following species.
- (i) BF_4^- (3 marks)
- (ii) NH_4^+ (3 marks)
- (b) Consider the molecule ammonia (NH_3).
- (i) Draw the electron diagram of a NH_3 molecule, showing electrons in the outermost shells only (1 mark)
- (ii) Draw the electron diagram of a BF_3 molecule, showing electrons in the outermost shells only.
(Hint: The B atom in BF_3 does NOT have an octet structure). (1 mark)
- (iii) With reference to the electronic arrangement of boron trifluoride, explain why it is unstable. (1 mark)
- (iv) Describe the bond formation between ammonia and boron trifluoride. (1 mark)
- (v) Draw the structural formula of F_3BNH_3 . (1 mark)

35. When hydrogen chloride gas is passed into water, H_3O^+ ions will be formed

- (a) In each of the following boxes, draw the electron diagram (showing electrons in the outermost shells only) for a suitable chemical species to show the formation of a H_3O^+ ion.



(2 marks)

- (b) Hence, describe the formation of dative covalent bond using H_3O^+ as example.

(1 mark)

- (c) The formation of H_3O^+ involves the formation of dative covalent bond.

- (i) What is meant by the term 'dative covalent bond'?

(1 mark)

- (ii) Draw the structural formula of a H_3O^+ ion.

(1 mark)

- (iii) What is the chemical name of H_3O^+ ion?

(1 mark)

36. Both carbon monoxide and nitric acid have a dative covalent bond in their own molecules.

(a) Below is the electron 'dot-and-cross' diagram for a molecule of carbon monoxide.



(i) How many bond pairs of electrons are there in a carbon monoxide molecule?

(1 mark)

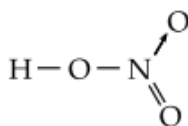
(ii) How many lone pairs of electrons are there in a carbon monoxide molecule?

(1 mark)

(iii) Write the structural formula for a molecule of carbon monoxide. Use an arrow ($\rightarrow$) to represent any dative covalent bond formed.

(1 mark)

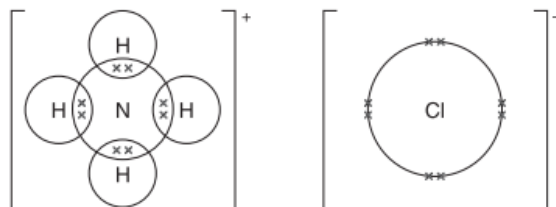
(b) The diagram below shows the bonding in one nitric acid molecule.



Draw the electron diagram for a nitric acid molecule, showing electrons in the outermost shells only. Use crosses to represent the electron of the H atom; dots to represent the electrons of the O atoms and small circles for the electrons of the N atom.

(1 mark)

19. The electron diagram (only electrons in the outermost shells are shown.) of ammonium chloride is shown below:



(a) Identify the type of bond which exists between the two ions.

(1 mark)

(b) Identify the type of bond which exists between nitrogen and hydrogen.

(1 mark)

(c) When a hydrogen ion reacts with an ammonia molecule, an ammonium ion is formed.

(i) Give the name of the type of bond formed when a hydrogen ion reacts with an ammonia molecule.

(1 mark)

(ii) Describe how this bond is formed in the ammonium ion.

(1 mark)

GROUP 族

I		II		relative atomic mass																atomic number		原子序																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
3	Li 6.9	4	Be 9.0	5	B 10.8	6	C 12.0	7	N 14.0	8	O 16.0	9	F 19.0	10	Ne 20.2	11	Na 23.0	12	Mg 24.3	13	Al 27.0	14	Si 28.1	15	P 31.0	16	S 32.1	17	Cl 35.5	18	Ar 40.0	19	K 39.1	20	Ca 40.1	21	Sc 45.0	22	Ti 47.9	23	V 50.9	24	Cr 52.0	25	Mn 54.9	26	Fe 55.8	27	Co 58.9	28	Ni 58.7	29	Cu 63.5	30	Zn 65.4	31	Ga 69.7	32	Ge 72.6	33	As 74.9	34	Se 79.0	35	Br 79.9	36	Kr 83.8	37	Rb 85.5	38	Sr 87.6	39	Y 88.9	40	Zr 91.2	41	Nb 92.9	42	Mo 95.9	43	Tc (98)	44	Ru 101.1	45	Rh 102.9	46	Pd 106.4	47	Ag 107.9	48	Cd 112.4	49	In 114.8	50	Sn 118.7	51	Sb 121.8	52	Te 127.6	53	I 126.9	54	Xe 131.3	55	Cs 132.9	56	Ba 137.3	57 *	La 138.9	58	Ce 140.1	59	Pr 140.9	60	Nd 144.2	61	Pm (144.9)	62	Sm 150.4	63	Eu 151.9	64	Gd 157.3	65	Tb 158.9	66	Dy 162.5	67	Ho 164.9	68	Er 167.3	69	Tm 168.9	70	Yb 173.0	71	Lu 175.0	72	Hf 178.5	73	Ta 180.9	74	W 183.9	75	Re 186.2	76	Os 190.2	77	Ir 192.2	78	Pt 195.1	79	Au 197.0	80	Hg 200.6	81	Tl 204.4	82	Pb 207.2	83	Bi 209.0	84	Po (209)	85	At (210)	86	Rn (222)	87	Fr (223)	88	Ra (226)	89 **	Ac (227)	90	Th (232)	91	Pa (231)	92	U 238.0	93	Np (237)	94	Pu (244)	95	Am (243)	96	Cm (247)	97	Bk (247)	98	Cf (251)	99	Es (252)	100	Fm (257)	101	Mn (288)	102	Sg (293)	103	Bh (310)	104	Hs (310)	105	Ds (310)	106	Uut (315)	107	Uuq (315)	108	Uub (315)	109	Uut (315)	110	Uuq (315)	111	Uub (315)	112	Uut (315)	113	Uuq (315)	114	Uub (315)	115	Uut (315)	116	Uuq (315)	117	Uub (315)	118	Uut (315)	119	Uuq (315)	120	Uub (315)	121	Uut (315)	122	Uuq (315)	123	Uub (315)	124	Uut (315)	125	Uuq (315)	126	Uub (315)	127	Uut (315)	128	Uuq (315)	129	Uub (315)	130	Uut (315)	131	Uuq (315)	132	Uub (315)	133	Uut (315)	134	Uuq (315)	135	Uub (315)	136	Uut (315)	137	Uuq (315)	138	Uub (315)	139	Uut (315)	140	Uuq (315)	141	Uub (315)	142	Uut (315)	143	Uuq (315)	144	Uub (315)	145	Uut (315)	146	Uuq (315)	147	Uub (315)	148	Uut (315)	149	Uuq (315)	150	Uub (315)	151	Uut (315)	152	Uuq (315)	153	Uub (315)	154	Uut (315)	155	Uuq (315)	156	Uub (315)	157	Uut (315)	158	Uuq (315)	159	Uub (315)	160	Uut (315)	161	Uuq (315)	162	Uub (315)	163	Uut (315)	164	Uuq (315)	165	Uub (315)	166	Uut (315)	167	Uuq (315)	168	Uub (315)	169	Uut (315)	170	Uuq (315)	171	Uub (315)	172	Uut (315)	173	Uuq (315)	174	Uub (315)	175	Uut (315)	176	Uuq (315)	177	Uub (315)	178	Uut (315)	179	Uuq (315)	180	Uub (315)	181	Uut (315)	182	Uuq (315)	183	Uub (315)	184	Uut (315)	185	Uuq (315)	186	Uub (315)	187	Uut (315)	188	Uuq (315)	189	Uub (315)	190	Uut (315)	191	Uuq (315)	192	Uub (315)	193	Uut (315)	194	Uuq (315)	195	Uub (315)	196	Uut (315)	197	Uuq (315)	198	Uub (315)	199	Uut (315)	200	Uuq (315)	201	Uub (315)	202	Uut (315)	203	Uuq (315)	204	Uub (315)	205	Uut (315)	206	Uuq (315)	207	Uub (315)	208	Uut (315)	209	Uuq (315)	210	Uub (315)	211	Uut (315)	212	Uuq (315)	213	Uub (315)	214	Uut (315)	215	Uuq (315)	216	Uub (315)	217	Uut (315)	218	Uuq (315)	219	Uub (315)	220	Uut (315)	221	Uuq (315)	222	Uub (315)	223	Uut (315)	224	Uuq (315)	225	Uub (315)	226	Uut (315)	227	Uuq (315)	228	Uub (315)	229	Uut (315)	230	Uuq (315)	231	Uub (315)	232	Uut (315)	233	Uuq (315)	234	Uub (315)	235	Uut (315)	236	Uuq (315)	237	Uub (315)	238	Uut (315)	239	Uuq (315)	240	Uub (315)	241	Uut (315)	242	Uuq (315)	243	Uub (315)	244	Uut (315)	245	Uuq (315)	246	Uub (315)	247	Uut (315)	248	Uuq (315)	249	Uub (315)	250	Uut (315)	251	Uuq (315)	252	Uub (315)	253	Uut (315)	254	Uuq (315)	255	Uub (315)	256	Uut (315)	257	Uuq (315)	258	Uub (315)	259	Uut (315)	260	Uuq (315)	261	Uub (315)	262	Uut (315)	263	Uuq (315)	264	Uub (315)	265	Uut (315)	266	Uuq (315)	267	Uub (315)	268	Uut (315)	269	Uuq (315)	270	Uub (315)	271	Uut (315)	272	Uuq (315)	273	Uub (315)	274	Uut (315)	275	Uuq (315)	276	Uub (315)	277	Uut (315)	278	Uuq (315)	279	Uub (315)	280	Uut (315)	281	Uuq (315)	282	Uub (315)	283	Uut (315)	284	Uuq (315)	285	Uub (315)	286	Uut (315)	287	Uuq (315)	288	Uub (315)	289	Uut (315)	290	Uuq (315)	291	Uub (315)	292	Uut (315)	293	Uuq (315)	294	Uub (315)	295	Uut (315)	296	Uuq (315)	297	Uub (315)	298	Uut (315)	299	Uuq (315)	300	Uub (315)	301	Uut (315)	302	Uuq (315)	303	Uub (315)	304	Uut (315)	305	Uuq (315)	306	Uub (315)	307	Uut (315)	308	Uuq (315)	309	Uub (315)	310	Uut (315)	311	Uuq (315)	312	Uub (315)	313	Uut (315)	314	Uuq (315)	315	Uub (315)	316	Uut (315)	317	Uuq (315)	318	Uub (315)	319	Uut (315)	320	Uuq (315)	321	Uub (315)	322	Uut (315)	323	Uuq (315)	324	Uub (315)	325	Uut (315)	326	Uuq (315)	327	Uub (315)	328	Uut (315)	329	Uuq (315)	330	Uub (315)	331	Uut (315)	332	Uuq (315)	333	Uub (315)	334	Uut (315)	335	Uuq (315)	336	Uub (315)	337	Uut (315)	338	Uuq (315)	339	Uub (315)	340	Uut (315)	341	Uuq (315)	342	Uub (315)	343	Uut (315)	344	Uuq (315)	345	Uub (315)	346	Uut (315)	347	Uuq (315)	348	Uub (315)	349	Uut (315)	350	Uuq (315)	351	Uub (315)	352	Uut (315)	353	Uuq (315)	354	Uub (315)	355	Uut (315)	356	Uuq (315)	357	Uub (315)	358	Uut (315)	359	Uuq (315)	360	Uub (315)	361	Uut (315)	362	Uuq (315)	363	Uub (315)	364	Uut (315)	365	Uuq (315)	366	Uub (315)	367	Uut (315)	368	Uuq (315)	369	Uub (315)	370	Uut (315)	371	Uuq (315)	372	Uub (315)	373	Uut (315)	374	Uuq (315)	375	Uub (315)	376	Uut (315)	377	Uuq (315)	378	Uub (315)	379	Uut (315)	380	Uuq (315)	381	Uub (315)	382	Uut (315)	383	Uuq (315)	384	Uub (315)	385	Uut (315)	386	Uuq (315)	387	Uub (315)	388	Uut (315)	389	Uuq (315)	390	Uub (315)	391	Uut (315)	392	Uuq (315)	393	Uub (315)	394	Uut (315)	395	Uuq (315)	396	Uub (315)	397	Uut (315)	398	Uuq (315)	399	Uub (315)	400	Uut (315)	401	Uuq (315)	402	Uub (315)	403	Uut (315)	404	Uuq (315)	405	Uub (315)	406	Uut (315)	407	Uuq (315)	408	Uub (315)	409	Uut (315)	410	Uuq (315)	411	Uub (315)	412	Uut (315)	413	Uuq (315)	414	Uub (315)	415	Uut (315)	416	Uuq (315)	417	Uub (315)	418	Uut (315)	419	Uuq (315)	420	Uub (315)	421	Uut (315)	422	Uuq (315)	423	Uub (315)	424	Uut (315)	425	Uuq (315)	426	Uub (315)	427	Uut (315)	428	Uuq (315)	429	Uub (315)	430	Uut (315)	431	Uuq (315)	432	Uub (315)	433	Uut (315)	434	Uuq (315)	435	Uub (315)	436	Uut (315)	437	Uuq (315)	438	Uub (315)	439	Uut (315)	440	Uuq (315)	441	Uub (315)	442	Uut (315)	443	Uuq (315)	444	Uub (315)	445	Uut (315)	446	Uuq (315)	447	Uub (315)	448	Uut (315)	449	Uuq (315)	450	Uub (315)	451	Uut (315)	452	Uuq (315)	453	Uub (315)	454	Uut (315)	455	Uuq (315)	456	Uub (315)	457	Uut (315)	458	Uuq (315)	459	Uub (315)	460	Uut (315)	461	Uuq (315)	462	Uub (315)	463	Uut (315)	464	Uuq (315)	465	Uub (315)	466	Uut (315)	467	Uuq (315)	468	Uub (315)	469	Uut (315)	470	Uuq (315)	471	Uub (315)	472	Uut (315)	473	Uuq (315)	474	Uub (315)	475	Uut (315)	476	Uuq (315)	477	Uub (315)	478	Uut (315)	479	Uuq (315)	480	Uub (315)	481	Uut (315)	482	Uuq (315)	483	Uub (315)	484	Uut (315)	485	Uuq (315)	486	Uub (315)	487	Uut (315)	488	Uuq (315)	489	Uub (315)	490	Uut (315)	491	Uuq (315)	492	Uub (315)	493	Uut (315)	494	Uuq (315)	495	Uub (315)	496	Uut (315)	497	Uuq (315)	498	Uub (315)	499	Uut (315)	500	Uuq (315)	501	Uub (315)	502	Uut (315)	503	Uuq (315)	504	Uub (315)	505	Uut (315)	506	Uuq (315)	507	Uub (315)	508	Uut (315)	509	Uuq (315)	510	Uub (315)	511	Uut (315)	512	Uuq (315)	513	Uub (315)	514	Uut (315)	515	Uuq (315)	516	Uub (315)	517	Uut (315)	518	Uuq (315)	519	Uub (315)	520	Uut (315)	521	Uuq (315)	522	Uub (315)	523	Uut (315)	524	Uuq (315)	525	Uub (315)	526	Uut (315)	527	Uuq (315)	528	Uub (315)	529	Uut (315)	530	Uuq (315)	531	Uub (315)	532	Uut (315)	533	Uuq (315)	534	Uub (315)	535	Uut (315)	536	Uuq (315)	537	Uub (315)	538	Uut (315)	539	Uuq (315)	540	Uub (315)	541	Uut (315)	542	Uuq (315)	543	Uub (315)	544	Uut (315)	545	Uuq (315)	546	Uub (315)	547	Uut (315)	548	Uuq (315)	549	Uub (315)	550	Uut (315)	551	Uuq (315)	552	Uub (315)	553	Uut (315)	554	Uuq (315)	555	Uub (315)	556	Uut (315)	557	Uuq (315)	558	Uub (315)	559	Uut (315)	560	Uuq (315)	561	Uub (315)	562	Uut (315)	563	Uuq (315)	564	Uub (315)	565	Uut (315)	566	Uuq (315)	567	Uub (315)	568	Uut (315)	569	Uuq (315)	570	Uub (315)	571	Uut (315)	572	Uuq (315)	573	Uub (315)	574	Uut (315)	575	Uuq (315)	576	Uub (315)	577	Uut (315)	578	Uuq (315)	579	Uub (315)	580	Uut (315)	581	Uuq (315)	582	Uub (315)	583	Uut (315)	584	Uuq (315)	585	Uub (315)	586	Uut (315)	587	Uuq (315)	588	Uub (315)	589	Uut (315)	590	Uuq

●

58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu
140.1		140.9		144.2		(145)		150.4		152.0		157.3		158.9		162.5		164.9		167.3		168.9		173.0		175.0	
90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk	98	Cf	99	Es	100	Fm	101	Md	102	No	103	Lr
232.0		(231)		238.0		(237)		(244)		(243)		(247)		(247)		(251)		(252)		(257)		(258)		(259)		(260)	

Part 5

1. Aluminium is a Group III metal.

- (a) With the aid of a labelled diagram, briefly discuss the formation of metallic bond in aluminium. Hence, explain why aluminium has a high melting point

(2 marks)

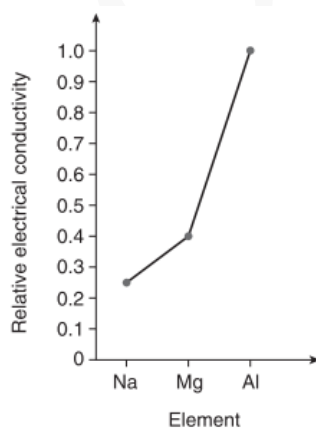
- (b) Solid aluminium is a good conductor of electricity. Consider the following sentences:

- In a piece of aluminium, **particle X** move freely and randomly in all directions. When both ends of the metal piece are connected to the terminals of a battery, **particle X** flow in one direction only towards the **terminal Y** of the battery. At the same time, an equal number of **particle X** flow into the other end of the metal from the opposite terminal of the battery.

What are **particle X** and **terminal Y**?

(2 marks)

2. The diagram below shows the relative electrical conductivities of the first three elements in Period 3.



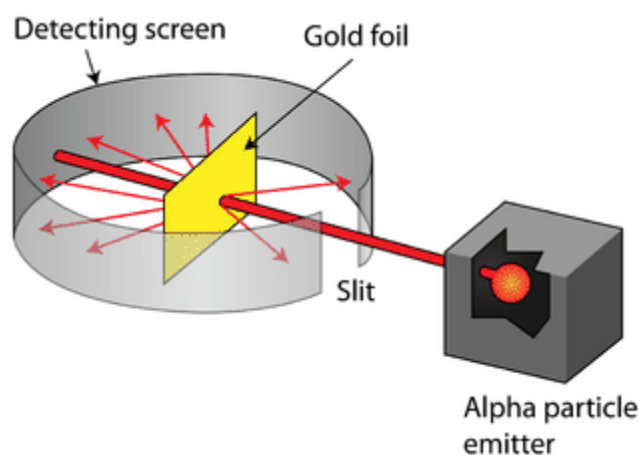
- (a) Explain why the electrical conductivity of aluminium is higher than that of magnesium.

(2 marks)

- (b) Explain why metals are ductile.

(2 marks)

3. In 1911, Ernest Rutherford and his coworkers performed the gold foil scattering experiment. In the experiment, they bombarded very thin sheets of gold foil with fast moving **alpha particles**. Alpha particles can be represented by the symbol ${}^4\text{He}^{2+}$.

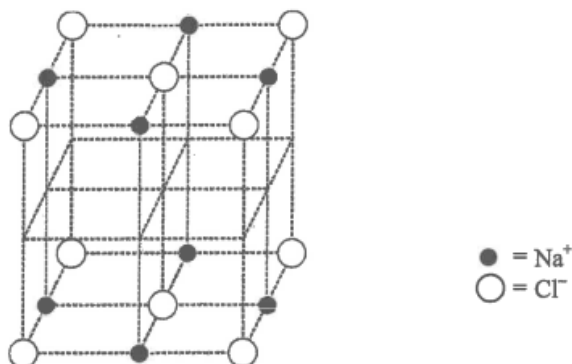


From the experimental result, most of the alpha particles passed through without deflection. However, a very small percentage bounced off at very large angles. Rutherford suggested that the atom has a very small positively charged region with a dense mass at the center.

- (a) Explain why only a very small percentage of alpha particles were deflected and bounced off at very large angles. (2 marks)
- (b) What are the numbers of proton, neutron and electron in an alpha particle? (1 mark)
- (c) Radium (atomic number = 88) was used as the source of alpha particle in the emitter. It is known that both radium and calcium can react with dilute hydrochloric acid to give metal chloride. Draw the electron diagram of radium chloride, showing electrons in the outermost shells only. (1 mark)
- (d) In terms of structure and bonding, explain why a piece of gold can be hammered down to foil. (2 marks)

4. Both sodium chloride (NaCl) and caesium chloride (CsCl) are ionic compounds with giant ionic structure.

(a) The diagram below shows a part of the structure of sodium chloride crystal with some ions missing.



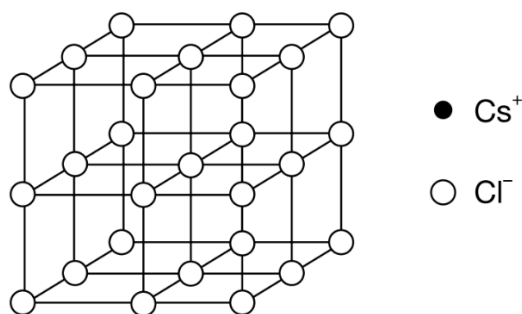
(i) Complete the diagram by using ● as Na^+ ion and ○ as Cl^- ion.

(1 mark)

(ii) Explain why sodium chloride has a higher melting point.

(1 mark)

(b) The diagram below shows a part of structure of caesium chloride crystal with some ions missing



(i) Complete the diagram by using ● as Cs^+ ion.

(1 mark)

(ii) Explain why CsCl(s) cannot NOT conduct electricity.

(1 mark)

(c) Which metal, sodium or caesium, has a higher melting point?

(1 mark)

(d) Which metal, sodium or caesium, reacts with chlorine more vigorously? Explain your answer.

(1 mark)

5. Sodium and caesium (Cs) are both alkali metals. They can react with iodine to form metal iodide.

(a) Explain why they are described as 'alkali'.

(1 mark)

(b) Which metal reacts with iodine more vigorously? Explain your answer.

(1 mark)

(c) The structure of sodium iodide is very similar to that of sodium chloride. Draw a diagram to show the structure of sodium iodide.

(2 marks)

(d) The structure of caesium iodide is very similar to that of caesium chloride. Draw a diagram to show the structure of caesium iodide.

(2 marks)

(e) Sodium iodide is soluble in water but insoluble in organic solvents like hexane. Complete the following sentences concerning the explanation.

- When a sodium iodide crystal is added to water, attraction exists between _____ in sodium iodide and water molecules. This attraction causes the _____ and _____ to move away from the crystal and go into the water.
- No such attraction exists between the _____ in sodium iodide and molecules of hexane. Thus, sodium iodide is insoluble in hexane.

(3 marks)

7. Rubidium is an alkali metal. It has two naturally occurring isotopes, ^{85}Rb and ^{87}Rb . The relative abundance of ^{85}Rb is 2.6 times greater than that of ^{87}Rb .
- (a) (i) What is meant by the term 'relative atomic mass'? (1 mark)
- (ii) Calculate the relative atomic mass of rubidium. (2 marks)
- (b) Rubidium reacts vigorously with chlorine to give rubidium chloride.
- (i) Draw the electron diagram for rubidium chloride, showing electrons in the outermost shells only. (1 mark)
- (ii) State one similarity and one difference in the structures of rubidium and rubidium chloride. (2 marks)
- (iii) State and explain the difference in electrical conductivity of rubidium and rubidium chloride in the solid state. (2 marks)

8. Many scientists contributed to the development of the periodic table. One of these was Dmitri Mendeléeve. In his table, he arranged the elements in order of increasing relative atomic mass. In the modern periodic table, elements are arranged in order of increasing atomic number.

(a) Explain the term 'atomic number'.

(1 mark)

(b) When the elements are arranged in order of increasing relative atomic mass, iodine would appear in Group VI. Give ONE reason why iodine must appear in Group VII.

(1 mark)

(c) Gallium is the element below aluminium in the periodic table. When Mendeléeve produced his periodic table, gallium had not been discovered. He left a gap for gallium below aluminium and predicted the properties of gallium and its compounds. The table below shows Mendeléeve's predictions and the properties we now know gallium to have.

	Mendeléeve's predictions	Properties of gallium
Relative atomic mass	About 68	70
Melting point / °C	Low	30
Reaction of oxide with acids	Forms colorless solution	Forms colorless solution

(i) Give ONE piece of evidence from the table that shows that Mendeléeve's predictions were correct.

(1 mark)

(ii) Both aluminium oxide and gallium oxide are ionic compounds. The symbol for an atom of gallium is Ga. Suggest the formula of gallium oxide.

(1 mark)

(ii) Calculate the formula mass of aluminium oxide, Al_2O_3 . (Relative atomic masses: O = 16.0, Al = 27.0)

(1 mark)

(iii) Predict ONE property of gallium oxide that is NOT given in the table.

(1 mark)

(d) Mendeléeve did not know of the existence of the noble gases. Helium and neon are noble gases and do not form any compounds. Why do helium and neon not form any compounds?

(1 mark)

9. Nitrogen is the most abundant element in air.

(a) Draw the electron diagram of a nitrogen molecule, showing electrons in outermost shell only.

(1 mark)

(b) In terms of structure and bonding, explain why nitrogen has a very low melting point.

(1 mark)

(c) Explain why liquid nitrogen is a poor conductor of electricity.

(1 mark)

(d) Ammonia is a compound formed between nitrogen and hydrogen.

(i) Draw the electron diagram of ammonia, showing electrons in outermost shell only.

(1 mark)

(ii) Explain why aqueous ammonia is a good conductor of electricity.

(1 mark)

(e) Nitrogen and neon are gases under room conditions. The boiling points of nitrogen and neon are tabulated below.

	Nitrogen	Neon
Boiling point (°C)	-195.8	-246.0

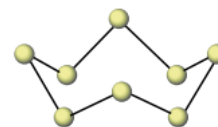
Explain, in terms of intermolecular forces, why the boiling point of nitrogen is higher than that of neon.

(2 marks)

10. The models below show the structure of a white phosphorus molecule and that of a sulphur molecule.



White phosphorus

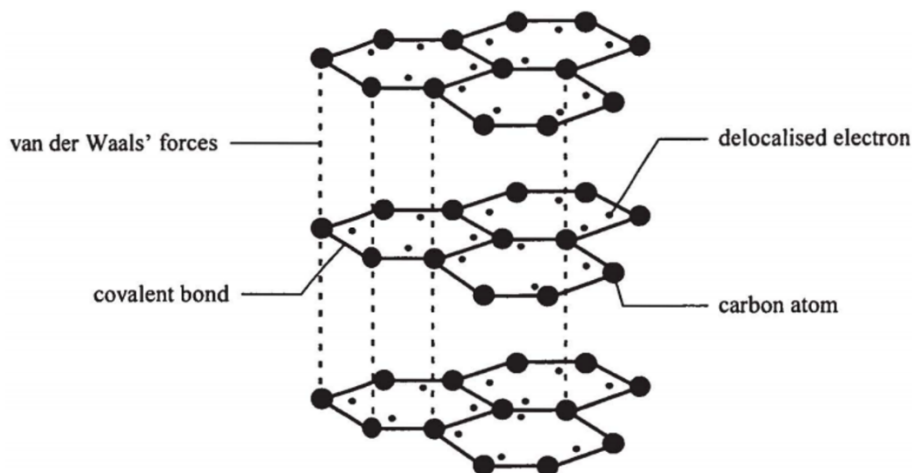


Sulphur

- (a) State the type of bonding that exists between the atoms in a white phosphorus molecule. (1 mark)
- (b) State the attraction that hold white phosphorus molecules together. (1 mark)
- (c) Explain why sulphur cannot conduct electricity. (1 mark)
- (d) White phosphorus is soluble in organic solvent like tetrachloromethane (CCl_4) but insoluble in water. Complete the following sentences concerning the explanation.
 - The attraction between water molecules is quite strong. The weak attraction between _____ and _____ molecules is not strong enough to overcome the attraction between the water molecules. Thus, white phosphorus molecules and water molecules cannot mix together. This explains why white phosphorus has a very low solubility in water
 - In organic solvents like tetrachloromethane, the molecules are held together by weak attraction. Thus, _____ molecules and _____ molecules can mix together more readily. (2 marks)
- (e) In terms of structure and bonding, explain why solid sulphur is brittle. (2 marks)
- (f) Calculate the relative molecular mass of white phosphorus. (Relative atomic mass of P = 31.0) (1 mark)

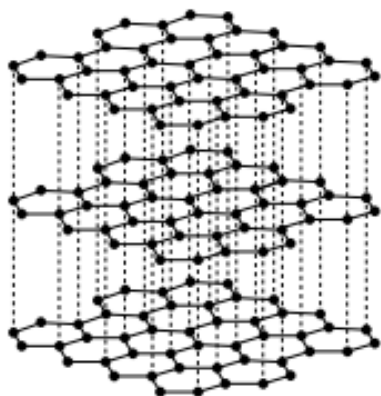
11. Diamond and graphite are different forms of carbon. Both of them have a giant covalent structure. However, they show differences in their properties.
- (a) Draw a diagram to show the structure of
- (i) diamond. (1 mark)
- (ii) graphite. (1 mark)
- (b) Explain why graphite can conduct electricity. (1 mark)
- (c) Diamond and graphite show different hardness. Explain why
- (i) diamond is an extremely hard substance; (1 mark)
- (ii) graphite is a soft substance. (1 mark)
- (d) Both diamond and graphite melt over 3000°C . Suggest why they have such high melting points. (1 mark)
- (e) Explain why graphite is insoluble in any solvent. (1 mark)

12. The structure of graphite is shown in the diagram below:

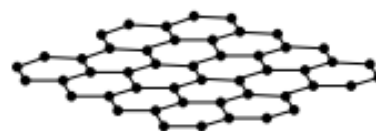


- (a) Name the structure that graphite belongs to. (1 mark)
- (b) Explain why graphite can be used to make solid lubricant (1 mark)
- (c) Explain why graphite has a very high melting point (1 mark)
- (d) Graphite is commonly used to make pencil cores for forming markings on paper. Tom claims that lead metal can also be used for forming markings.
 - (i) With the aid of a suitable diagram, describe the structure and bonding of lead. (3 marks)
 - (ii) Hence, explain why Tom's claim is INCORRECT. (1 mark)
- (e) Diamond is another form of carbon.
 - (i) Draw the structure of diamond (1 mark)
 - (ii) With reference to the structure and particles, explain why electrode is made of graphite, but NOT diamond (1 mark)

13. Each layer of carbon atoms in graphite is known to have a honeycomb structure. When a force is applied to graphite, individual layers of carbon atoms can be removed. A single layer of the carbon atoms from graphite is called 'graphene'.



structure of graphite



structure of graphene

- (a) Suggest why graphene can be separated from graphite when a force is applied. (1 mark)
- (b) What type of structure does graphene have? (1 mark)
- (c) Graphene is an excellent electrical conductor. It is 300 times more conducting than copper. Suggest why graphene can conduct electricity. (1 mark)
- (d) Graphene is a strong material. It is 300 times stronger than steel. Explain, with reference to its bonding and structure, why graphene is strong. (1 mark)
- (e) Graphane is a new material produced by reacting graphene with hydrogen. In graphane, each carbon atom is bonded to three other carbon atoms and one hydrogen atom. Would you expect graphane to be an electrical insulator? Explain your answer. (2 marks)

14. Ammonia borane (H_3NBH_3) is a covalent molecule formed between BH_3 and NH_3 .

(a) Describe the formation of dative covalent bond using H_3NBH_3 as example,

(3 marks)

(b) Upon heating, H_3NBH_3 will be decomposed into hydrogen and boron nitride (BN).

(i) The structure of boron nitride is similar to that of graphite. Draw the structure of one layer of solid boron nitride.

Given that B and N atoms are in alternate positions.

(2 marks)

(ii) Hence, explain why the melting point of boron nitride is much higher than that of ammonia.

(2 marks)

(c) Draw the structural formula of the following molecules with dative covalent bond:

(i) Aminoborane (H_2BNH_2)

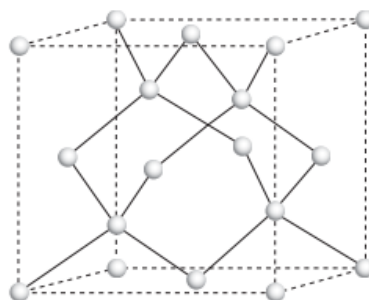
(1 mark)

(ii) Iminoborane (HBNH)

(1 mark)

15. Both carbon and silicon are elements in Group IV of the Periodic Table. Carbon dioxide and silicon dioxide are their oxides respectively.
- (a) Write the structural formula of carbon dioxide.
(1 mark)
 - (b) Draw a diagram to show how the atoms in silicon dioxide are bonded together.
(1 mark)
 - (c) Explain why carbon dioxide exists as a gas while silicon dioxide exists as a solid at room temperature and pressure.
(2 marks)
16. Calcium fluoride is an ionic compound. It can be made by reacting calcium with fluorine.
- (a) In terms of structure and bonding, explain why calcium fluoride has a high melting point.
(2 marks)
 - (b) The element fluorine exists as fluorine molecules, F_2 .
 - (i) Draw an electron diagram of a molecule of fluorine, showing outermost shell electrons only.
(1 mark)
 - (ii) Fluorine can be made into a solid by cooling it. Solid fluorine melts at $-220\text{ }^{\circ}\text{C}$. In terms of structure and bonding, explain why fluorine has such a low melting point.
(2 marks)

17. Silicon (Si) and germanium (Ge) have the same crystal structure, as shown below:



- (a) What type of structure do silicon and germanium have? (1 mark)
- (b) Silicon occurs in nature as silicon dioxide in sand and quartz. Draw the three-dimensional structure of silicon dioxide. (1 mark)
- (c) Silicon combines with chlorine to form silicon tetrachloride. Draw an electron diagram of the chloride, showing electrons in the outermost shells only. (1 mark)
- (d) Explain why silicon dioxide has a higher melting point than silicon tetrachloride in terms of their structures. (2 marks)
- (e) The table below gives the relative abundance of each isotope in a sample of germanium, Ge.

Isotope	^{70}Ge	^{72}Ge	^{74}Ge
Relative abundance / %	24.4	32.4	43.2

What is the relative atomic mass of Ge?

(2 marks)

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18. The following table shows the formulae and the melting points of the oxides of some Period 3 elements.

Group number	I	II	III	IV	VI
Element	Na	Mg	Al	Si	S
Formula of the oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	SO ₂
Melting point of the oxide / °C	920	2830	2040	2950	-75

(a) Describe the change in the classification of the elements across the period.

(1 mark)

(b) Which of these oxides has/have a

(i) giant ionic structure?

(1 mark)

(ii) giant covalent structure?

(1 mark)

(iii) simple molecular structure?

(1 mark)

(c) Explain why the melting point of magnesium oxide is much higher than that of sulphur dioxide.

(2 marks)

19. State and explain the difference between the electrical conductivity of sodium, iodine and sodium iodide.

(6 marks)

20. Consider three different chemical species below:

Ammonium chloride

Methane

Potassium

(a) Complete the following table.

	Ammonium chloride	Methane	Potassium
Type(s) of chemical bonding	1.	1.	1.
	2.	2.	

(5 marks)

(b) State and explain the electrical conductivities of the above three compounds in solid states

(4 marks)

21. W, X, Y and Z are four elements in the Periodic Table. The total sum of atomic numbers of X, Y and Z equals to 45. X and Z are both Group VII elements and the atomic number of Z is smaller than that of X. In addition, W, X and Y are three consecutive elements in the Periodic Table.

(a) What are elements W, X, Y and Z?

(4 marks)

(b) Compound A is formed from X and Y.

(i) Name the type of bonding formed in compound A.

(1 mark)

(ii) Draw the electron diagram of compound A, showing electrons in outermost shells only.

(1 mark)

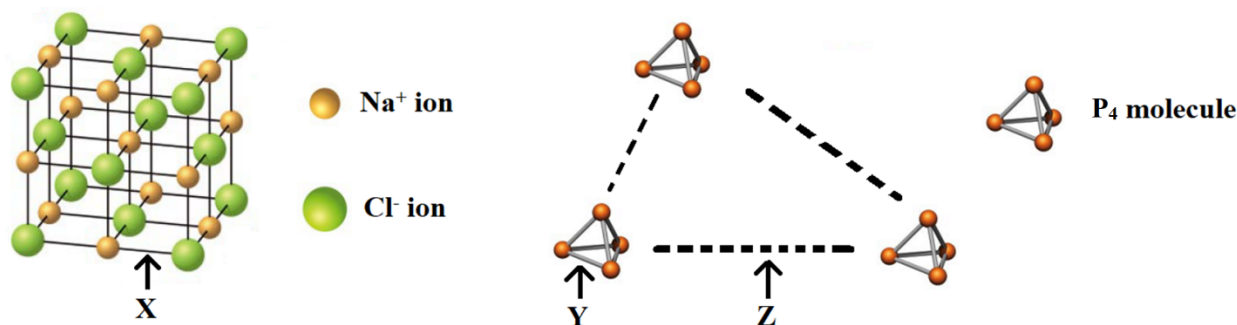
(c) Compare the electrical conductivity of compound A, element W and element Y in LIQUID STATE with reference to the type and property of the particles in it.

(3 marks)

(d) What will be observed, if any, when element W is heated strongly in air? Explain your answer.

(2 marks)

22. The diagram below shows a part of the structure of sodium chloride (NaCl) and phosphorus (P_4)



- (a) Name the types of attraction represented by X, Y and Z. (3 marks)
- (b) Explain why both substances CANNOT conduct electricity in solid state. (2 marks)
- (c) Which of the above substances ($NaCl$ and P_4) has a higher melting point? Explain your answer. (2 marks)

23. Iodine and graphite are both black solids. When iodine is heated gently, a purple vapour is seen. Graphite will not melt until the temperature reaches 4 000 K. Graphite conducts electricity but iodine is a very poor conductor of electricity.

- (a) State the type of crystal structure for each of iodine and graphite. (2 marks)
- (b) Explain why the melting point of graphite is very high. (1 mark)
- (c) Explain why iodine vaporizes easily when heated gently. (1 mark)
- (d) State why solid iodine is a very poor conductor of electricity. (1 mark)

24. Copper, graphite, diamond and silicon dioxide have different applications.
- (a) Suggest TWO properties of copper which make it suitable for making electric wires.
(2 marks)
 - (b) Graphite can conduct electricity. It is often used in making electrodes, but NOT electric wires.
 - (i) Suggest why graphite can conduct electricity.
(1 mark)
 - (ii) Suggest why graphite is NOT used in making electric wires.
(1 mark)
 - (c) Apart from making electrodes, graphite can be used as a lubricant. Explain, with reference to its bonding and structure, why graphite has lubricating property.
(1 mark)
 - (d) Diamond is very hard and can be used to make glass cutters. Explain, with reference to its bonding and structure, why diamond is so hard.
(1 mark)
 - (e) Silicon dioxide can be used as furnace lining, a material which covers and protects the interior surface of a furnace.
 - (i) Suggest a property of silicon dioxide which makes it suitable for making furnace lining.
(1 mark)
 - (ii) Explain how the structure of silicon dioxide relates to this property.
(1 mark)

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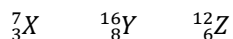
25. Boron and aluminium are Group III elements. They react with fluorine to give different fluorides. The physical properties of two fluorides are shown in the table below.

	Melting point (°C)	Boiling point (°C)	Electrical conductivity	
			Solid state	Molten state
BF₃	-127	-100	?	Poor
AlF₃	1290 (sublimes)	---	?	Good

- (a) State the physical state of the following compounds at room temperature.
- (i) BF₃ (1 mark)
- (ii) AlF₃ (1 mark)
- (b) Give the chemical name of the following compounds
- (i) BF₃ (1 mark)
- (ii) AlF₃ (1 mark)
- (c) Draw the electron diagrams, showing electrons in the outermost shell only, of the following compounds.
- (i) BF₃ (1 mark)
- (ii) AlF₃ (1 mark)
- (d) State, with explanation, whether solid BF₃ and solid AlF₃ can conduct electricity. (2 marks)
- (e) Describe the formation of dative covalent bond between BF₃ and ammonia. (3 marks)

26. Chlorine reacts with calcium and nitrogen differently to form different compounds. Compound X is formed by the reaction between calcium and chlorine while compound Y is formed by the reaction between nitrogen and chlorine.
- (a) Consider compound X:
- (i) Draw the electron diagram of compound X.
(1 mark)
 - (ii) Give the chemical name of compound X.
(1 mark)
 - (iii) Does compound X have a high melting point? Explain your answer in terms of structure and bonding of compound X.
(2 marks)
- (b) Consider compound Y:
- (i) Draw the electron diagram of compound Y.
(1 mark)
 - (ii) Give the chemical name of compound Y.
(1 mark)
 - (iii) Does compound Y have a high melting point? Explain your answer in terms of structure and bonding of compound Y.
(2 marks)
- (c) Compare and explain the difference in electrical conductivity of compound X and compound Y.
(2 marks)

27. Consider the following 3 elements.



X reacts with Y to form compound A. Y reacts with Z to form compound B.

- (a) Draw the electron diagram for compound A, showing electrons in the outermost shells only. (1 mark)
- (b) Draw the structural formula of compound B. (1 mark)
- (c) Y itself can form a molecule Y_2 .
- (i) Draw the electron diagram of Y_2 . (1 mark)
- (ii) Name the type of attraction that can be found **within a Y_2 molecule**. (1 mark)
- (iii) Name the type of attraction that can be found **between Y_2 molecules**. (1 mark)
- (iv) Compare and explain whether Y_2 or X has a higher melting point in terms of the structure and bonding of both chemical species. (2 marks)
- (d) Compare and explain the difference in electrical conductivity of compound A and compound B. (2 marks)

28. The following table shows some information about the atoms of 4 elements

Element	Number of protons
W	3
X	7
Y	9
Z	15

Compound P is formed when W reacts with X. Compound Q is formed when Y reacts with Z.

- Draw an electron diagram for a molecule of element X, showing electrons in the outermost shell only. (1 mark)
- Draw the electron diagram of compound P, showing electrons in the outermost shell only. (1 mark)
- Draw the structural formula of compound Q. (1 mark)
- X and Y combine to form a covalent compound with a molecular formula X_2Y_2 . Draw an electron diagram of X_2Y_2 . (Showing electrons in the outermost shell only) (1 mark)
- Compare and explain the difference in melting point between compound P and compound Q. (2 marks)
- Compare and explain the difference in electrical conductivity between solid element W and solid compound Q. (2 marks)

29. Fluorine is a halogen. It can react with potassium and nitrogen respectively.

(a) State the type of compound formed between

(i) fluorine and potassium.

(1 mark)

(ii) fluorine and nitrogen.

(1 mark)

(b) Write the formula of the compound formed between

(i) fluorine and potassium.

(1 mark)

(ii) fluorine and nitrogen.

(1 mark)

(c) Deduce whether the compound formed between fluorine and potassium or that between fluorine and nitrogen has a higher melting point.

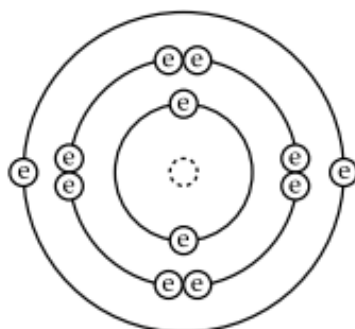
(1 mark)

(d) Complete the following table by filling in your predictions about other physical properties of these compounds.

	Compound formed between potassium and fluorine	Compound formed between nitrogen and fluorine
Solubility in water	High / Low	High / Low
Solubility in organic solvent	High / Low	High / Low
Electrical conductivity in solid state	Good / Poor	Good / Poor
Electrical conductivity in liquid state	Good / Poor	Good / Poor

(2 marks)

30. The electron diagram of an atom of element X is shown below.



Keys:

○ = nucleus

ⓔ = electron

- (a) State the number of protons that an atom of X has. (1 mark)
- (b) To which group of the Periodic Table does X belong? Explain your answer. (1 mark)
- (c) To which period of the Periodic Table does X belong? Explain your answer. (1 mark)
- (d) X has a giant metallic structure. Draw a diagram to show the bonding in the solid state of X. (1 mark)
- (e) The oxide of X is a white solid. Draw the electron diagram of this oxide, showing electrons in the outermost shells only. (1 mark)
- (f) The oxide of X has a melting point of over 2000°C. With reference to its structure and bonding, explain why this oxide has a high melting point. (2 marks)
- (g) The oxide of X has the same structure as sodium chloride. Draw a diagram to show the structure of the oxide of X. (1 mark)

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31. (a) Write down the type of forces of attraction (ionic bond / covalent bond / metallic bond / van der Waals' forces) present in each of the following cases:

	Forces of attraction
Between carbon atoms in diamond	
Between carbon dioxide molecules in dry ice	
Between particles in calcium nitride	
Between particles in magnesium	
Between carbon and oxygen atoms in a carbon dioxide molecule	

(5 marks)

- (b) Consider the following liquids:

molten potassium

molten sulphur

molten potassium chloride

liquid hydrogen chloride

- (i) Classify each of the above liquids in one of the following three ways according to its response to a direct current:

(4 marks)

Non-conductor	Conductor, with decomposition at the electrodes	Conductor, but without decomposition

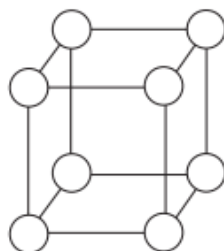
(3 marks)

- (ii) For each of the above liquids which are conductors, state the particles responsible for the conduction of electricity.

(2 marks)

32. At room temperature, both sodium metal and sodium chloride are crystalline solids.

(a) The diagram below shows part of the lattice of sodium chloride.



(i) In this diagram, mark all sodium ions with a positive (+) sign and all chloride ions with a negative (–) sign.

(1 mark)

(ii) What is the number of neighbouring chloride ions directly attached to each sodium ion in crystalline sodium chloride?

(1 mark)

(b) (i) Explain how the particles are held together in solid sodium metal.

(1 mark)

(ii) Explain how the particles are held together in solid sodium chloride.

(1 mark)

(iii) The melting point of sodium chloride is much higher than that of sodium metal. What can be deduced from this information?

(1 mark)

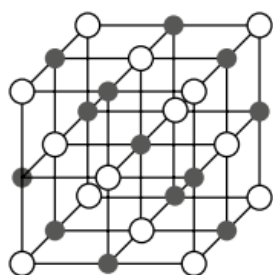
(c) Compare the electrical conductivity of solid sodium metal with that of solid sodium chloride. Explain your answer.

(2 marks)

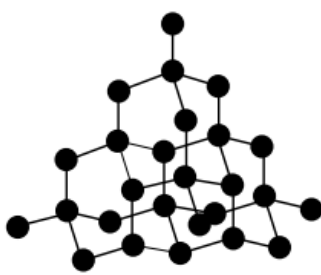
(d) Explain why sodium is malleable.

(2 marks)

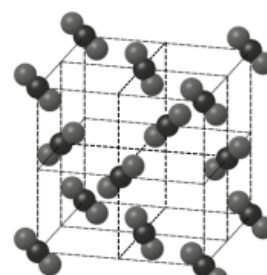
33. The structures of three solids X to Z are shown below. They can be diamond, sodium chloride and carbon dioxide.



X



Y



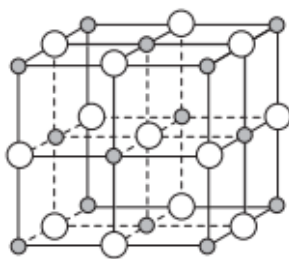
Z

- (a) Identify X, Y and Z respectively. (3 marks)
- (b) What types of structures do X, Y and Z have respectively? (3 marks)
- (c) Explain whether or not solid X can conduct electricity. (1 mark)
- (d) State ALL types of electrostatic attraction present in solid Z. (2 marks)
- (e) Explain whether Y or Z has a higher melting point. (2 marks)

34. Consider the substances listed below:

argon carbon dioxide hydrogen chloride iodine silicon dioxide sodium chloride

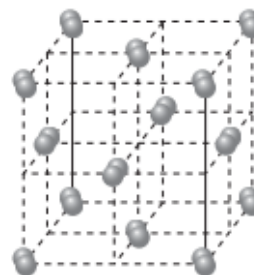
The following diagram shows the arrangements of particles in three substances X, Y and Z.



substance X



substance Y



substance Z

- (a) For each case, choose ONE substance from the above list which has arrangement of particles as shown under room temperature and pressure.

Substance X	
Substance Y	
Substance Z	

(3 marks)

- (b) Name the attractive forces that exist between particles in substance Y.

(1 mark)

- (c) Predict and explain the following properties of substance X:

(i) hardness;

(2 marks)

(ii) electrical conductivity in solid and molten states.

(2 marks)

- (d) Predict and the following properties of substance Z:

(i) solubility in water;

(1 mark)

(ii) solubility in non-aqueous solvents;

(1 mark)

(iii) electrical conductivity.

(1 mark)

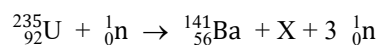
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35. Element X and element Y are elements in the same period. They can react with fluorine to form different fluoride. The physical properties of two fluorides are shown in the table below.

	Melting point (°C)	Boiling point (°C)	Electrical conductivity	
			Solid state	Liquid state
XF ₃	-151	-102	Poor	Poor
YF ₂	1263	2260	Poor	Good

- (a) State the physical state of the following compounds at room temperature.
- (i) XF₃ (1 mark)
- (ii) YF₂ (1 mark)
- (b) Name the structures adopted by the following compounds.
- (i) XF₃ (1 mark)
- (ii) YF₂ (1 mark)
- (c) Explain why the melting point of XF₃ is much lower than that of YF₂ (2 marks)
- (d) Explain the difference in electrical conductivity of YF₂ in solid state and liquid state (2 marks)
- (e) Given that ALL particles in YF₂ have the same electronic arrangement. Given the chemical name of the following compounds.
- (i) XF₃ (1 mark)
- (ii) YF₂ (1 mark)

36. On August 6, 1945, an atomic bomb was dropped on Hiroshima. This bomb exploded with an energy of approximately 15000 tons of trinitrotoluene (a kind of explosive). By bombarding neutrons to uranium-235, it undergoes nuclear fission with huge amount of energy released. The following equation represents one of the nuclear fission reactions of uranium-235. (Hint: ${}_0^1\text{n}$ in the equation below represents neutron)



- (a) What is the number of neutrons in ${}^{235}\text{U}$? (1 mark)
- (b) Uranium has three naturally occurring isotopes, ${}^{234}\text{U}$, ${}^{235}\text{U}$ and ${}^{238}\text{U}$. The nuclear fuel in the atomic bomb should be concentrated with ${}^{235}\text{U}$ by isotope separation. Explain why these isotopes CANNOT be separated by chemical means. (1 mark)
- (c) It is given that the numbers of neutrons and protons remain unchanged in the above nuclear fission reaction. Write down the FULL ATOMIC SYMBOL of X (2 marks)
- (d) Uranium can react with oxygen and fluorine to form uranium dioxide and uranium hexafluoride respectively. It is given that uranium dioxide melts at 2865°C while uranium hexafluoride sublimates at 56°C .
- (i) What is meant by the term 'sublime'? (1 mark)
- (ii) Name the type of structure for each of the following compounds:
- (1) Uranium dioxide (1 mark)
- (2) Uranium hexafluoride (1 mark)
- (iii) Explain whether solid uranium hexafluoride can conduct electricity. (1 mark)
- (iv) Explain whether liquid uranium dioxide can conduct electricity. (1 mark)

37. The relative atomic mass of silicon is 28.09. The table below lists the stable isotopes of silicon and the relative abundance of each.

Isotope	Abundance (%)
^{28}Si	x
^{29}Si	y
^{30}Si	3.1

- (a) What is meant by the term 'relative atomic mass'? (1 mark)
- (b) Calculate the values of x and y. (2 marks)
- (c) Account for the fact that CO_2 is a gas while SiO_2 is a high melting solid under room conditions. (2 marks)
- (d) The structures of elemental Si(s) and diamond are similar.
- (i) Draw a diagram to show the structure of silicon and indicate the interactions between the Si atoms. (1 mark)
- (ii) Hence, explain why silicon is hard whereas graphite is soft enough to be used as lubricant. (2 marks)
- (e) SiCl_4 exists as a liquid under normal conditions. In addition, $\text{SiCl}_4(\text{l})$ is UNABLE to conduct electricity. Explain the followings.
- (i) The boiling point of SiCl_4 is very low (57°C). (1 mark)
- (ii) $\text{SiCl}_4(\text{l})$ is a non-conductor. (1 mark)

PERIODIC TABLE 週期表

GROUP 族

atomic number 原子序

relative atomic mass 相對原子質量

		0															
		I		II										VII		0	
		3	4			21	22	23	24	25	26	27	28	29	30	5	2
		Li	Be			Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	B	He
		6.9	9.0			45.0	47.9	50.9	52.0	54.9	55.8	58.9	58.7	63.5	65.4	10.8	4.0
		11	12													13	10
		Na	Mg													Al	Ne
		23.0	24.3													27.0	20.2
		19	20			39	40	41	42	43	44	45	46	47	48	14	18
		K	Ca			Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Si	Ar
		39.1	40.1			88.9	91.2	92.9	95.9	(98)	101.1	102.9	106.4	107.9	112.4	28.1	40.0
		37	38													15	
		Rb	Sr													P	
		85.5	87.6													31.0	
		55	56			57 *	72	73	74	75	76	77	78	79	80	16	
		Cs	Ba			La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	S	
		132.9	137.3			138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	32.1	
		87	88			89 **	104	105								34	
		Fr	Ra			Ac	Rf	Db								Se	
		(223)	(226)			(227)	(261)	(262)								79.0	
																Br	
																35	
																53	
																I	
																126.9	
																85	
																At	
																(210)	
																86	
																Rn	
																(222)	

		58	59	60	61	62	63	64	65	66	67	68	69	70	71
		Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
		140.1	140.9	144.2	(145)	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
		90	91	92	93	94	95	96	97	98	99	100	101	102	103
		Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
		232.0	(231)	238.0	(237)	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

*

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